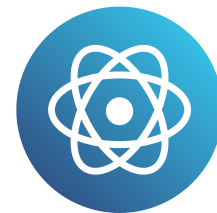
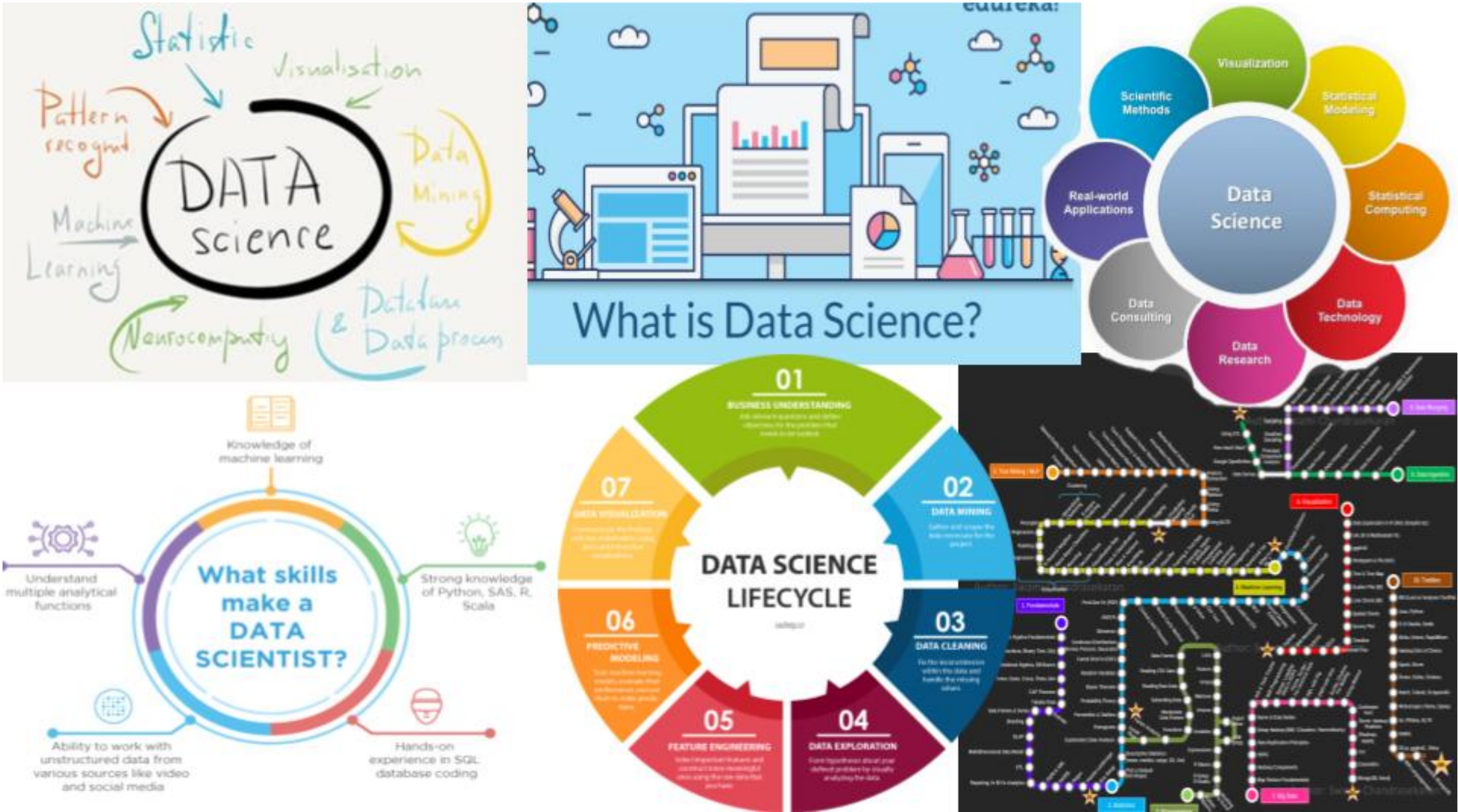


What is Data Science?

DATA SCIENCE FOR BUSINESS LEADERS

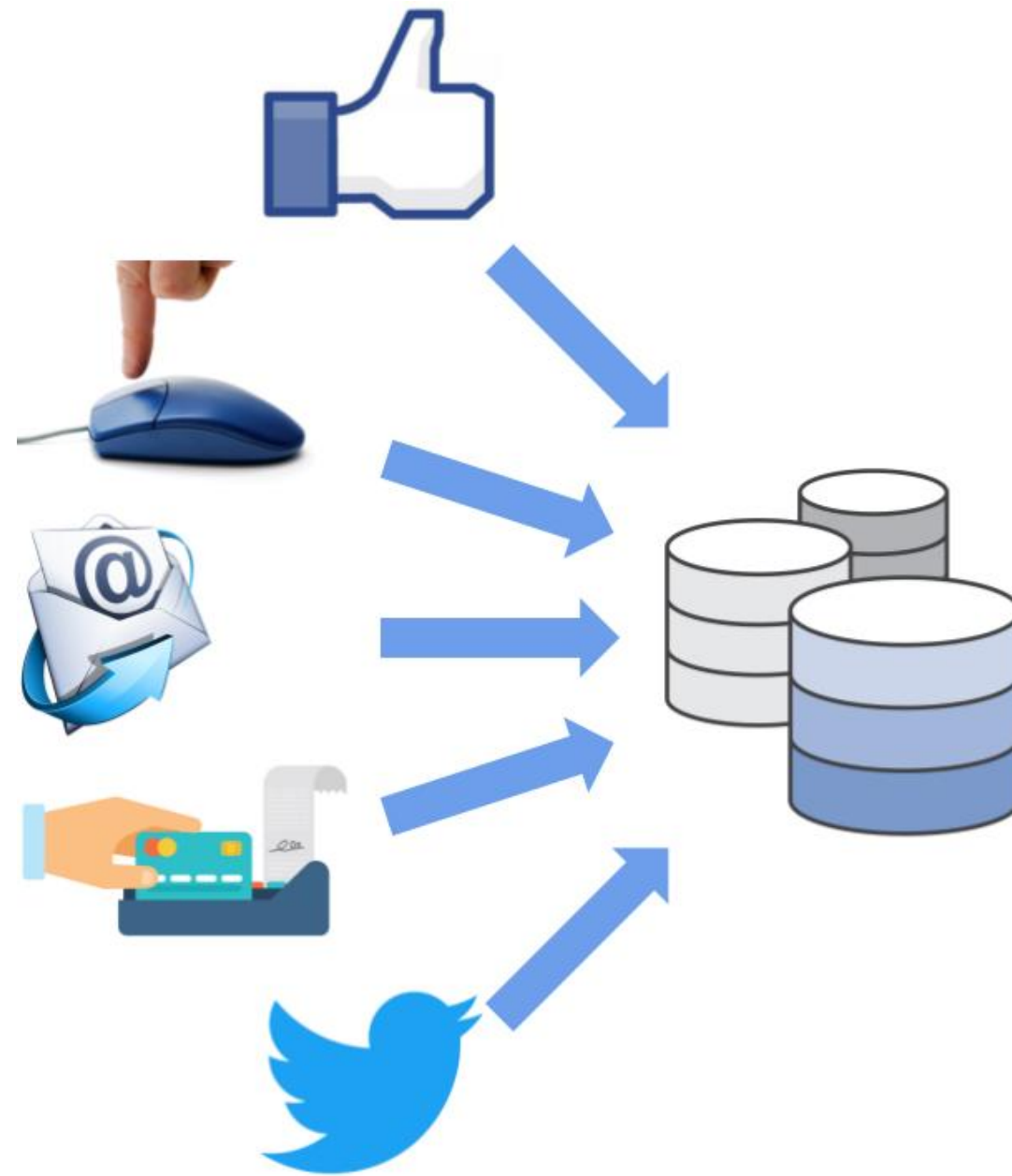


Let's ask Google!



Making data work for you

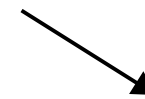
Data science is actually simple set of methodologies taking in thousands of forms of data that are available to us today and using them to draw meaningful conclusions.



Data is everywhere. Every like, click, Email, credit card swipe and Tweet Is a new piece of data that can be used to describe the present or better predict the future.

What can data do?

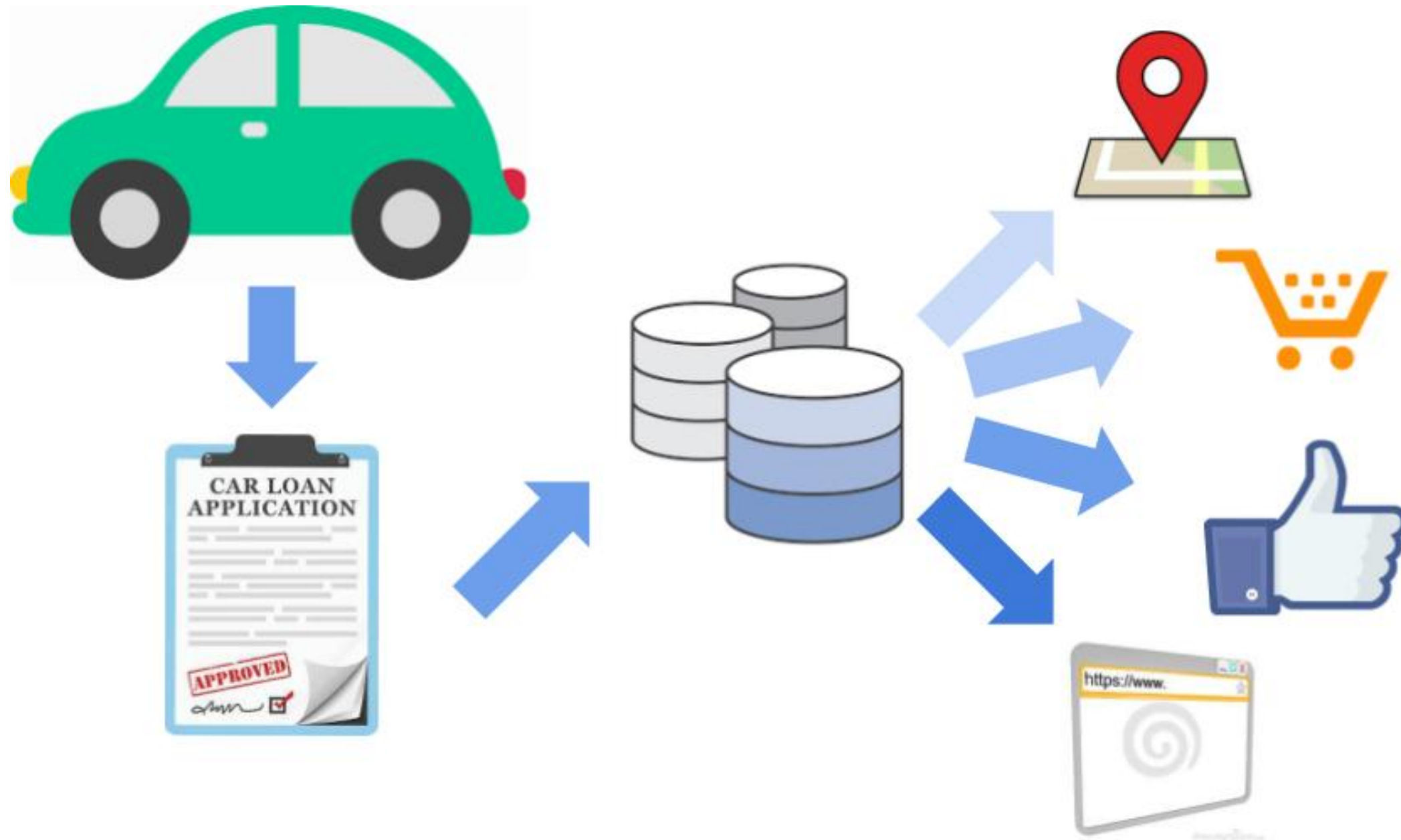
- Describe the current state of an organization or process
- Detect anomalous events
- Diagnose the causes of events and behaviors
- Predict future events



Data can describe a current state. This can be accomplished with dashboards or alerts, simplifying time-intensive reporting processes with new data Technologies.

Why now?

We are collecting more data than ever before. Suppose you fill out an application form for car loan. All of the data is automatically entered into a computer and combined with data from hundreds of dealerships into one with database.



Once we have that data it is easy to connect the email that you provided and tied with card purchase data from social media and web browsing. Now, we have a complete picture of everyone who purchased a car last year. This additional data can determine the price that you can pay for this car and helps a company to determine what other purchase you might make and how they can sell insurance in the best possible way.

The data science workflow

Data collection



Exploration and visualization



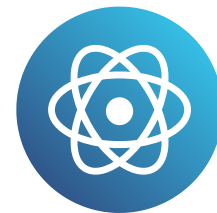
Experimentation and prediction



First, we collect data from various sources like Customer surveys, web traffic results, emails between a sales representative and potential clients and financial transactions. Next, We explore and visualize the data. this can help us explore and visualize our data and track how data changes over time. finally we make predictions with our data. for example, this can involve building a system that can segment clusters of customers or classify Pictures of different types of cars.

Applications of Data Science

DATA SCIENCE FOR BUSINESS LEADERS



Case study: fraud detection



Suppose you are the in=charge of Fraud detection at a large Bank. You might like to use data to determine the probability of transaction being Fake.

You might start with gathering information about each Purchase such as the amount, date and type of transaction made And cardholders address. We need many examples of transactions including this information as well as the label which tells you whether the transaction was fake or valid. you must have this data in your database which is known as training data and is used to build an algorithm. Every time a transaction is made, you give this data to the Algorithm. and it will predict whether the transaction was valid for fraudulent.

Case study: fraud detection

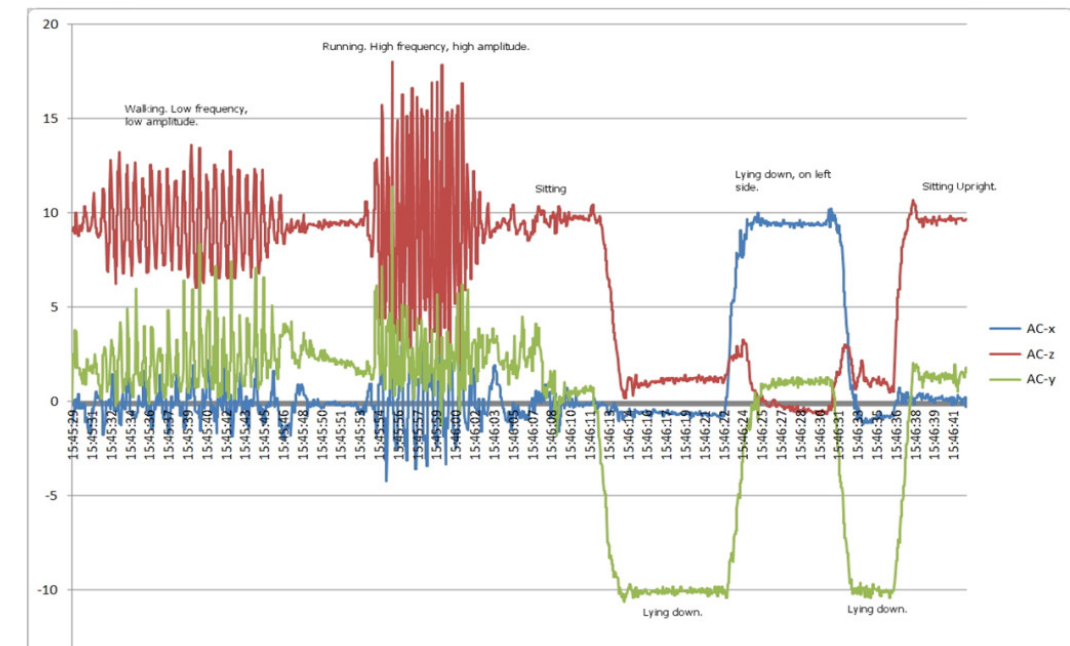


Amount	Date	Type	...
.	.	.	.
.	.	.	.
.	.	.	.
.	.	.	.
.	.	.	.
.	.	.	.
.	.	.	.

What do we need for machine learning?

- A well-defined question
 - *"What is the probability that this transaction is fraudulent?"*
- A set of example data
 - *Old transactions labeled as "fraudulent" or "valid"*
- A new set of data to use our algorithm on
 - *New credit card transactions*

Case study: smart watch



Internet of Things

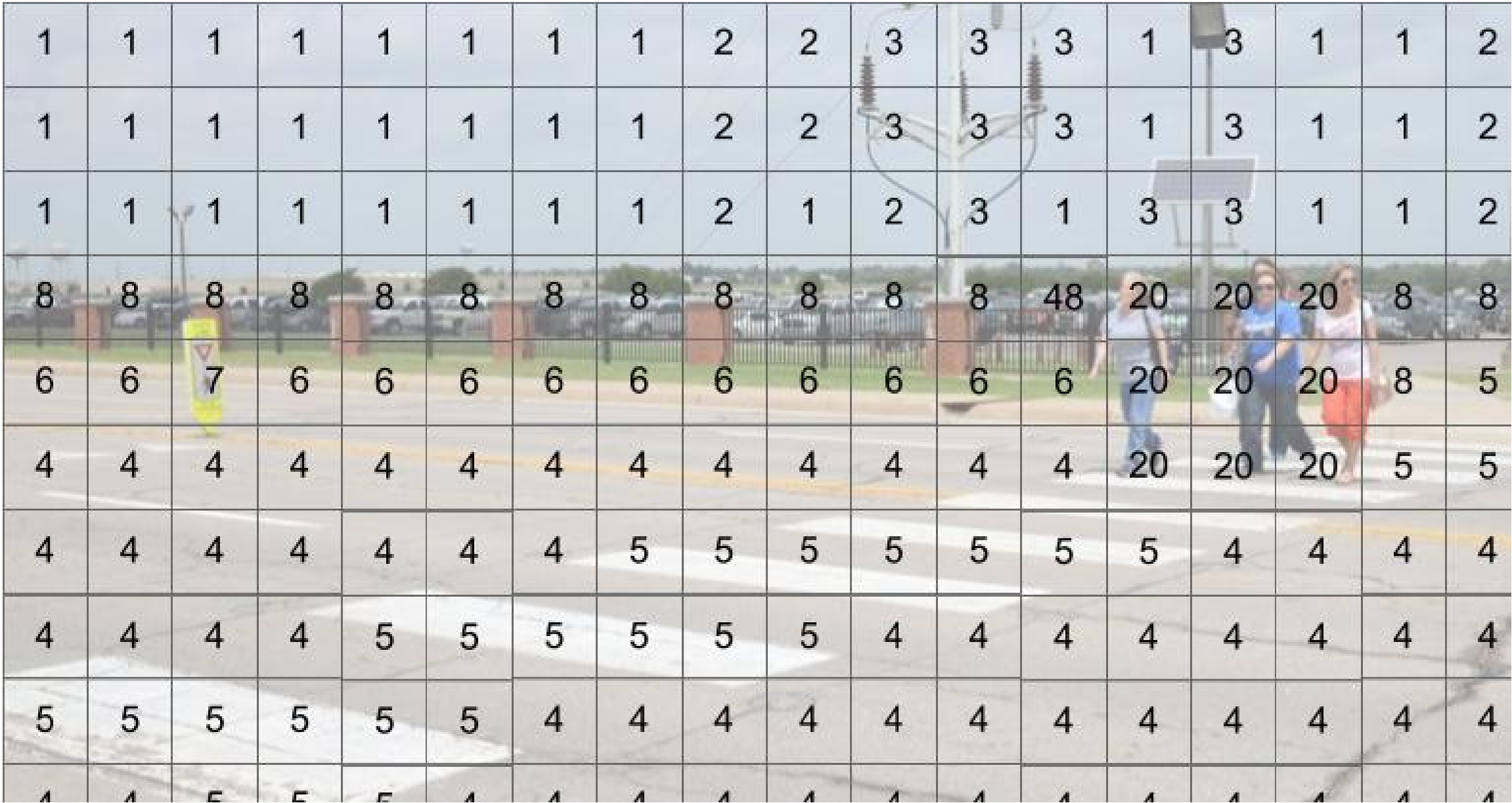
- Smart watches
- Internet-connected home security systems
- Electronic toll collection systems
- Building energy management systems
- Much, much more!

Suppose, you are trying to make a Smartwatch to monitor physical Activity. You want to be able to detect different physical activities like running or walking. A special sensor known as Accelerometer monitors motion in three dimensions. The data generated by the sensor is the basis of machine learning in the problem. you could ask several volunteers to record their activity when they are running or walking. Then, you can develop the algorithm that can recognize and accelerometer data as representing one of two States running or walking. It establishes a field which is known as internet of things which is also a part of data science. It talks about the Gadgets which are not standard computers but still have the ability to transmit data. this include smart watches, internet connected home security systems, stereo systems and fire alarm system.

Case study: image recognition



Case study: image recognition



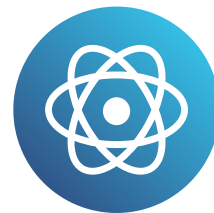
Deep learning

- Many neurons work together
- Requires much more training data
- Used in complex problems
 - Image classification
 - Language learning/understanding

Task for self driving Cars is to identify Image contains a human. It could express the pictures as a matrix of numbers which can act as a data set. However, this would feel if we feed this information into a model. for this we need advanced algorithms known as deep learning in which algorithms are built as neurons in multiple layers which work together to draw Complex conclusions. Deep learning takes much more data than a traditional machine learning model. it can also explore the relationships. It can solve Complex data problems like Image classification or language understanding.

Data sources and risks

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Common sources of data

- Web events
- Customer data
- Logistics data
- Financial transactions

Your company might already have collected this data. It's best to ask your data engineers what is collected and what is not.

Web data

- Events
- Timestamps
- User information

When a user visits a webpage or clicks on a link. It can be helpful to track this information in order to calculate conversion rate or monitor the popularity of different pieces of the content.

user_id	event_name	timestamp
1234	homepage_visit	2019-01-01 12:01:01

Personally Identifiable Information (PII)

Name	Timestamp	Object Clicked
Jane Doe	2019-01-20 12:05:00	Like Button

"Jane Doe" = Personally Identifiable Information (PII)

PII is any information that can help you track the identity of a human being like name, location, email address and phone Number

Data pseudonymization

user_id	Timestamp	Object Clicked
185477	2019-01-20 12:05:00	Like Button

user_id	Name
185477	Jane Doe

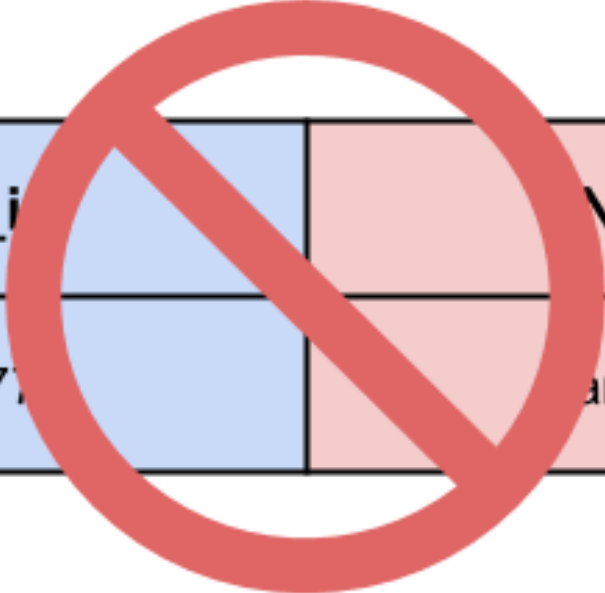
- Restricted access
- Audit logs

We can secure the PII of a person by splitting our table into two and allocating a user ID to the person and keeping the PII information in a separate table giving access to only authorized individuals. This is known as Data pseudonymization

Data anonymization

user_id	Timestamp	Object Clicked
185477	2019-01-20 12:05:00	Like Button

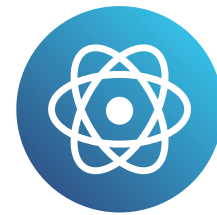
user_id	Name
185477	Jane Doe



The second option is to destroy PII after assigning a user ID to the person known as Data anonymization

Solicited data

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Why do we solicit data?

- Create marketing collateral
- De-risk decision making
- Monitor quality

Marketing Collateral:

A post about what percent of our customers are satisfied with our product.

De-Risk Decision Making:

When we survey users to gauge interest in a certain product

Monitor Quality:

Your institute asks the students to rate each course



Types of solicited data

- Surveys
- Customer reviews
- In-app questionnaires
- Focus groups

We appreciate your feedback!

X

Thank you for visiting our website. We are always looking for ways to improve your experience. Please take a moment to tell us about your experience.

How likely are you to recommend our website to a friend or colleague?

0

1

2

3

4

5

6

7

8

9

10

What could we do to improve your experience?

Send Feedback

powered by

 QuestionPro

Types of solicited data

Qualitative

- Conversations
- Open-ended questions

Subjective in nature and requires complex analysis.

Used to develop a hypotheses (a question to start research)

Quantitative

- Multiple choice
- Rating scale

Can be easily summarized in a chart or graph

Used to validate a hypotheses

Revealed and stated preferences

Stated preference :What people tell us they want or believe

- Hypothetical
- Subjective



Revealed preference :What their actions reveal

- Actions
- Purchasing decisions



Best practices

:To gain insights into revealed preference through stated preference

Be specific

Do this	Not that
On a scale of 1 - 5, how would you rate the quality of content on DataCamp?	How would you rate DataCamp?

Best practices

Be specific

Do this	Not that
On a scale of 1 - 5, how would you rate the quality of content on DataCamp?	How would you rate DataCamp?

Avoid loaded language

Do this	Not that
Which of the following political issues is most important to you?	Which of the following controversial political issues is most important to you?

Best practices

Calibrate By asking known quantities

Do this	Not that
Rate your interest in each of the following products at DataCamp.	Are you interested in Skill Assessment at DataCamp?

Best practices

Calibrate

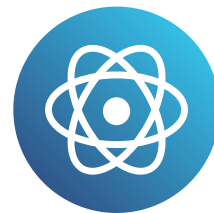
Do this	Not that
Are you interested in Skill Assessment at DataCamp?	Rate your interest in each of the following products at DataCamp.

Require actionable results

Do this	Not that
Have a hypothesis for each question	Ask a question just because it's interesting

Collecting additional data

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Internal sources of data are not enough. Often, you need to collect data from external sources as well

Even more data

- APIs
- Public records
- Mechanical Turk



Data APIs

- Application Programming Interface
- Request data over the internet
- Twitter
- Wikipedia
- Yahoo! Finance
- Google Maps
- Many more!

Tracking a hashtag

- All tweets with `#DataFramed` (DataCamp's podcast!)
- Use Twitter API

We have many options for analysis.

- We can perform sentiment analysis and get an idea of how people like our podcast.
- We can simply track how often `#data framed` appeared this week.
- We can also compare this data with download data to see how many hashtags have converted to download.

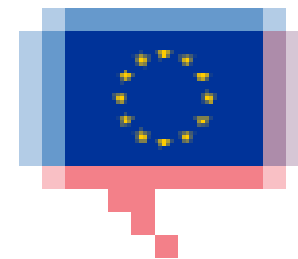


Public records

- For the US, data.gov
- For the EU, data.europa.eu

Data on literacy, education,
health and economic data

This is helpful in getting
population level trends

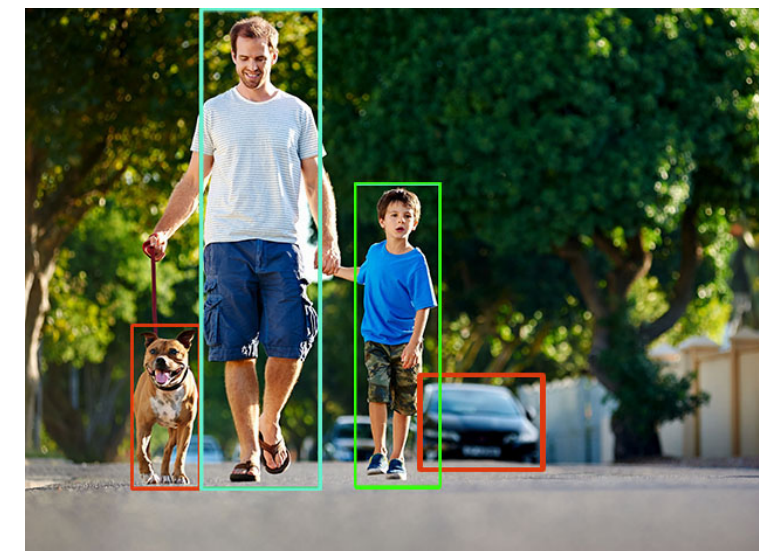
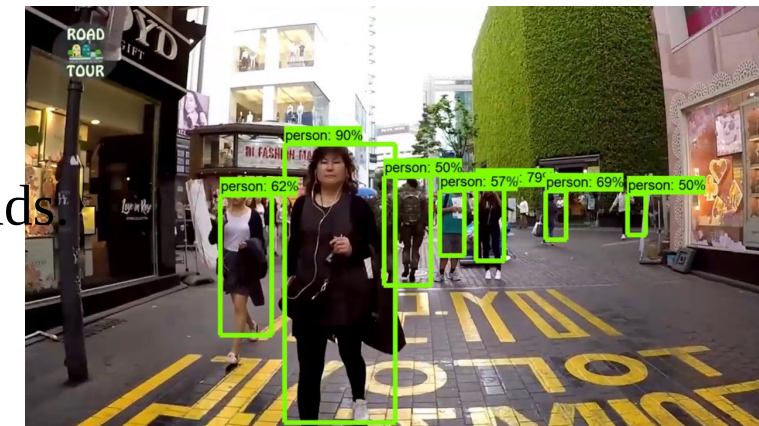


EU **Open Data** Portal

Building a training set



In order to develop training data, we need pictures with labels in thousands. Getting labeled images can be a difficult task.



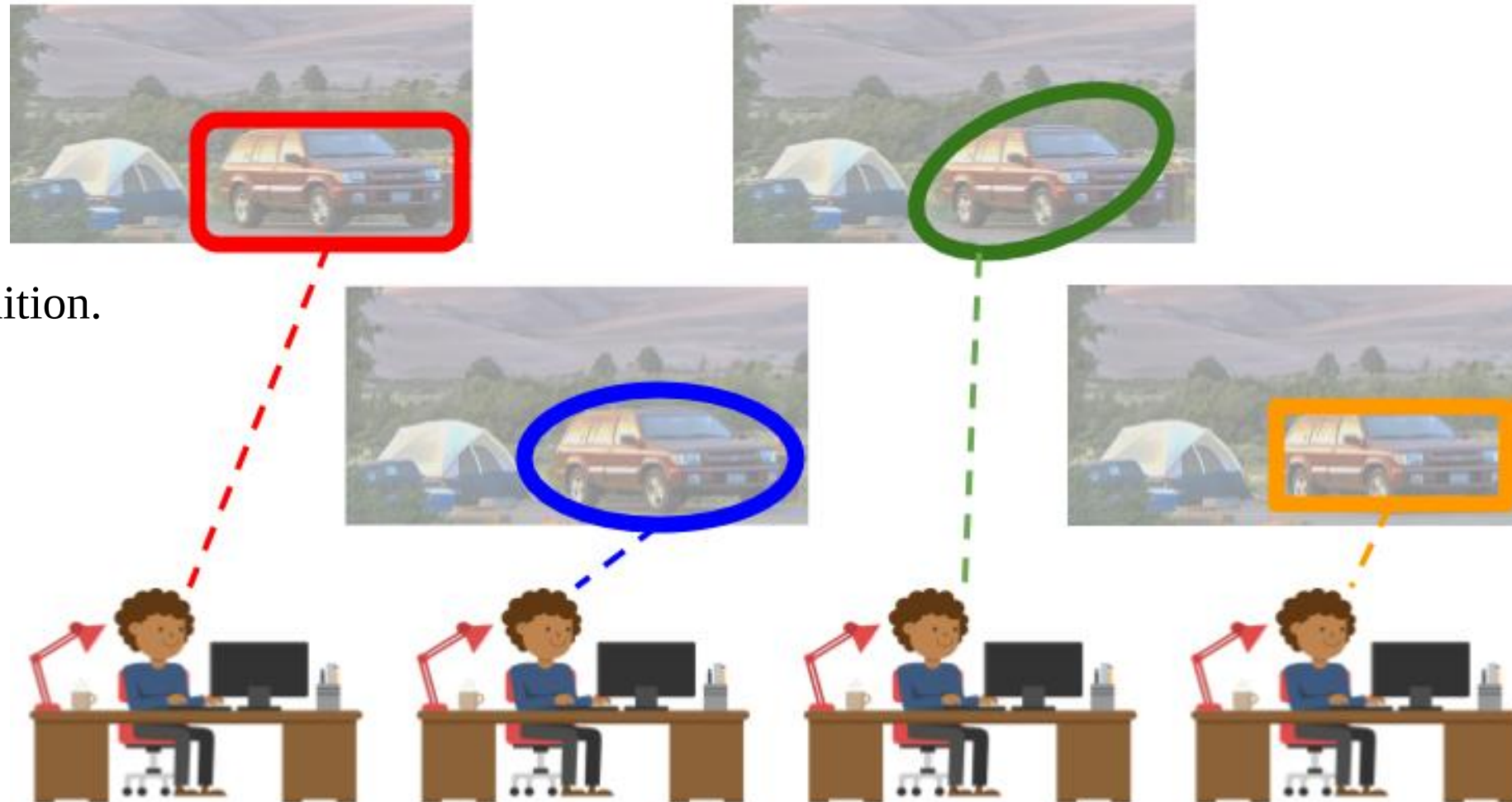
Mechanical Turk

Select the car in the image.

Asking humans to do a task which we plan to computerize.

In our previous example, this could mean labeling pictures for image recognition.

Rather than giving the task to one person, we get the task done by thousands of people



Mechanical Turk

:To ensure quality

- Resource: AWS MTurk
- Label customer reviews
- Extract text from a form
- Highlight key words in a sentence

Jane

Last Name

Smith

Email

stopall

Pick your color

☒ Red

☐ Green

☐

Submit

Select all squares with street signs.

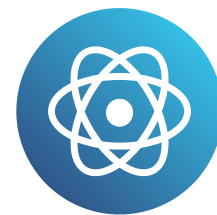


⌂ 🎧 ⓘ

VERIFY

Dashboards

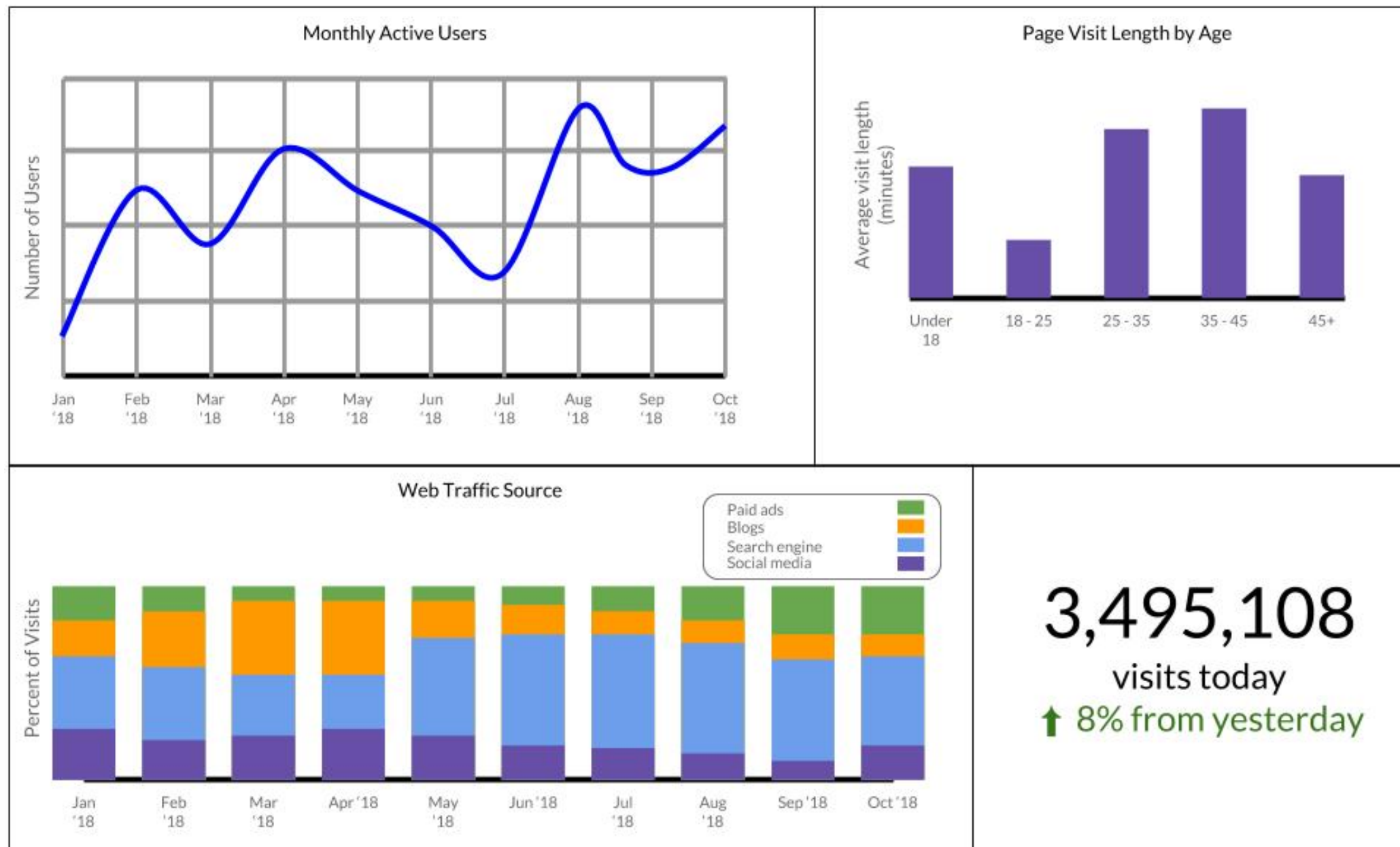
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What is a dashboard?

A dashboard is a set of metrics usually in the form of graphs that are updated on schedule.

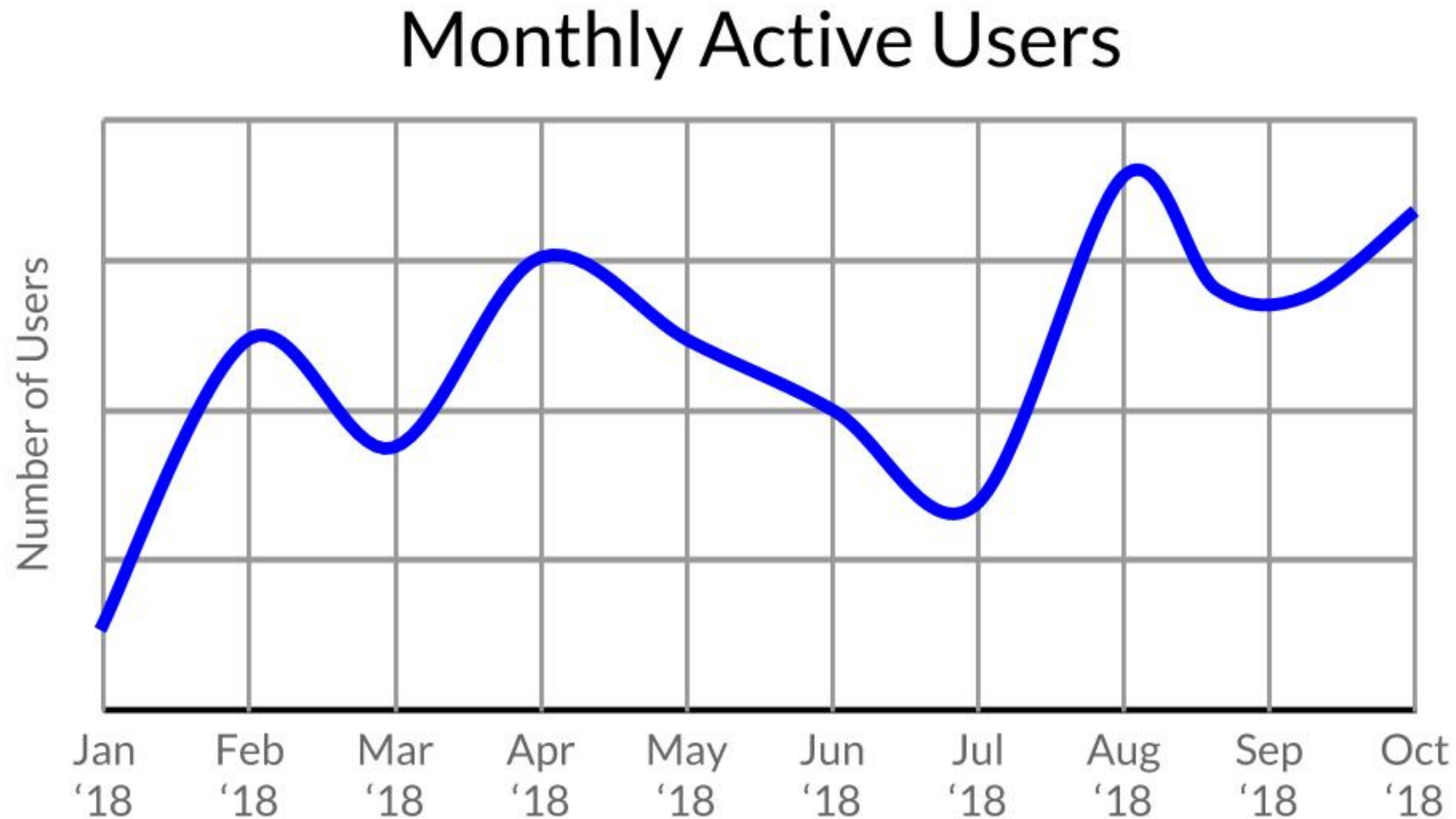
Some dashboards can update in real time as well.



Common Dashboard Elements

- Time Series
- Categorical Comparison
- Highlighting a single number

Tracking a value over time



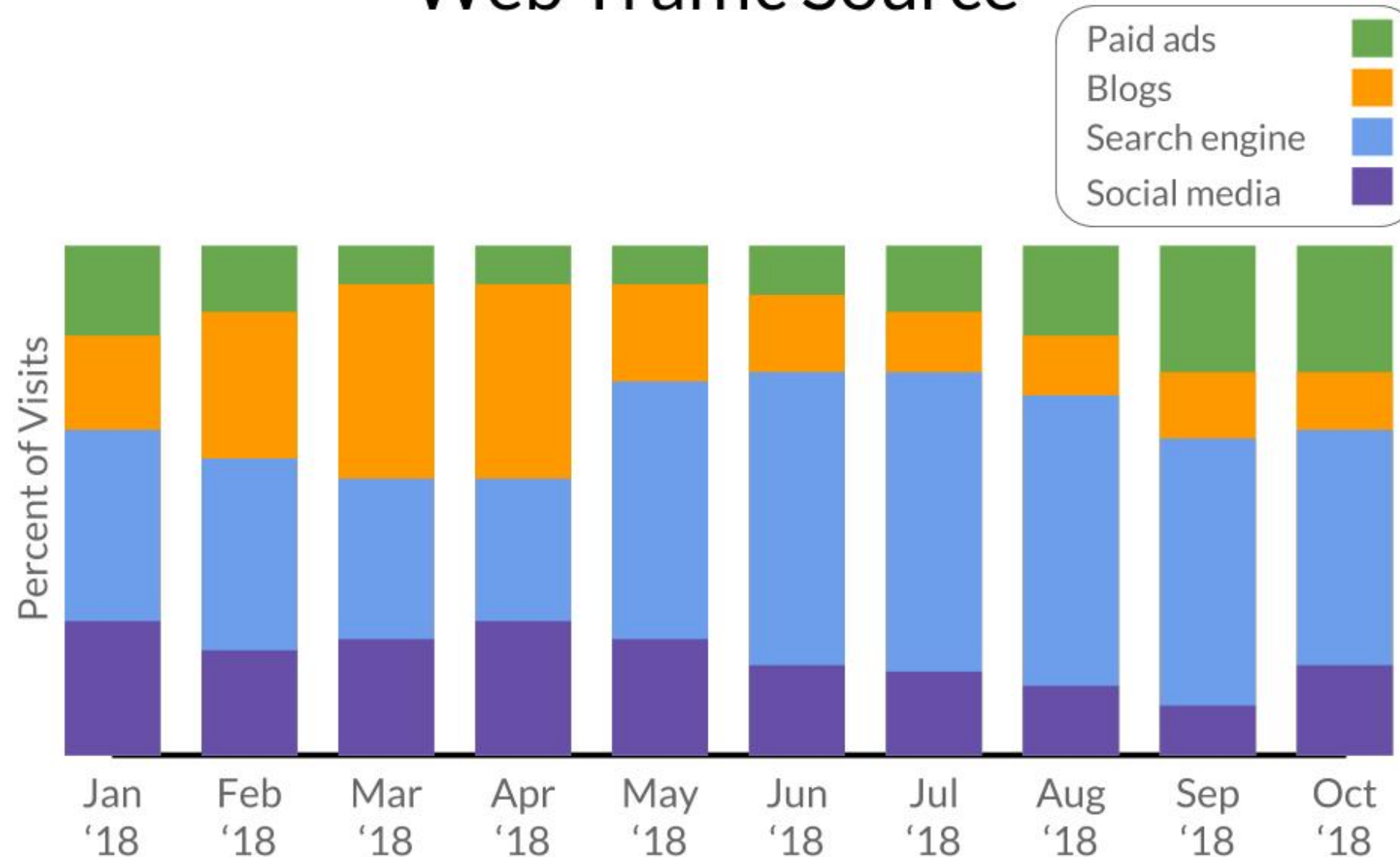
Tracking composition over time

Web Traffic Source

Give additional context to the previous chart.

It tells us the percent of users coming through different platforms

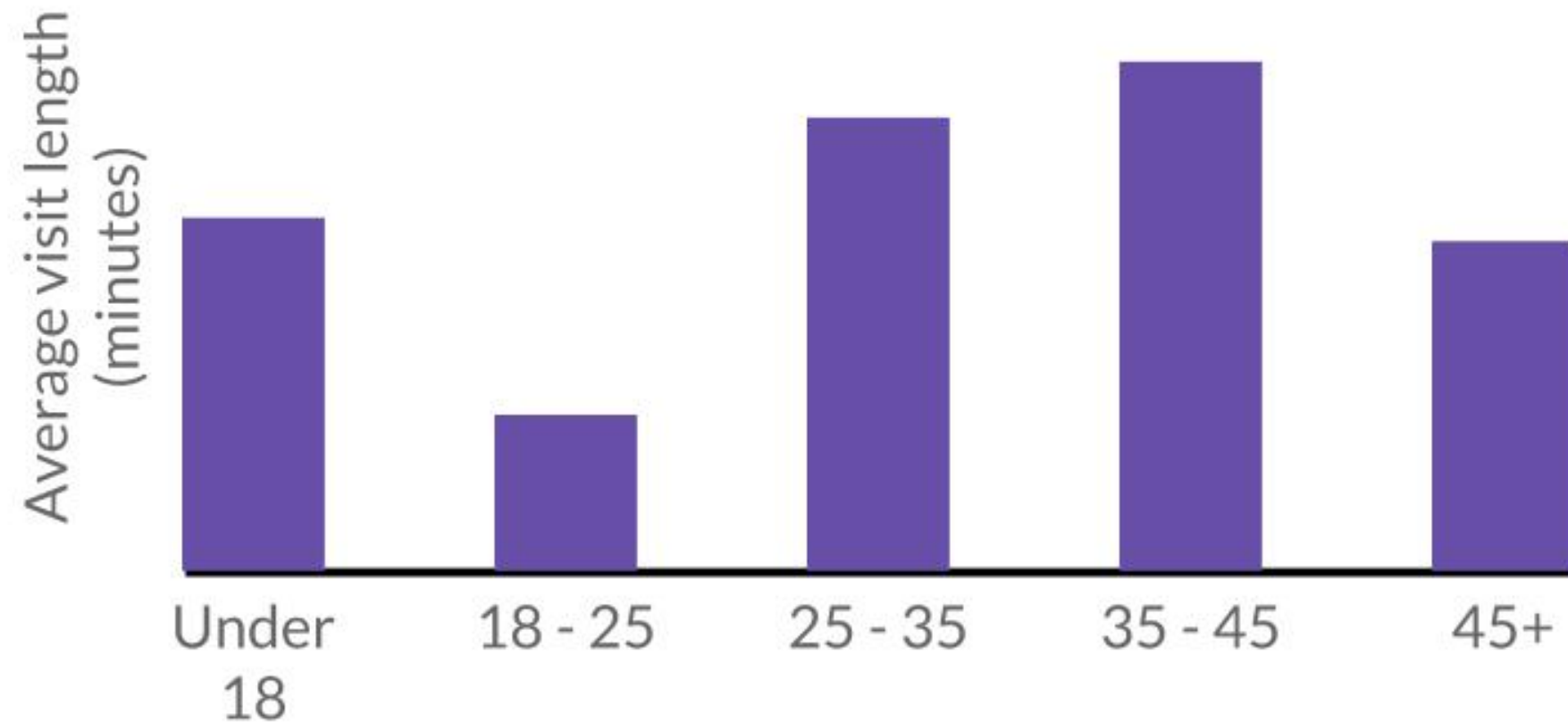
These are stacked bars and always total to 100 %



Categorical comparison

Bar Chart: Compares different groups during same time period

Page Visit Length by Age



Data from the past 30 days

Define time range
By adding label

Highlighting a single number

Can be used for making a figure prominent.

Can also be used to compare multiple values over time

3,495,108
visits today
↑ 8% from yesterday

Displaying text

Great way to add qualitative data like a few customer comment to signify the situation

Timestamp	Comment
Oct 9, 2018 12:57:05	Awesome website! Loved the new layout.
Oct 10 2018 03:16:00	Had trouble getting the website to load. I couldn't buy my favorite product!

Where can we build a dashboard?

- Spreadsheets: Excel or Google Sheets
- BI Tools: Power BI, Tableau, Looker
- Customized tools: R Shiny or d3.js

Be consistent across an organization!

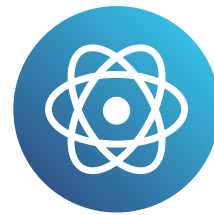
Requesting a dashboard

Is a dashboard the correct solution?

- Will you use it multiple times?
 - Does it need to be updated daily or weekly?
 - Will the request always be the same?
- Be specific
 - Specify your use case

Ad hoc analysis

DATA SCIENCE FOR BUSINESS LEADERS



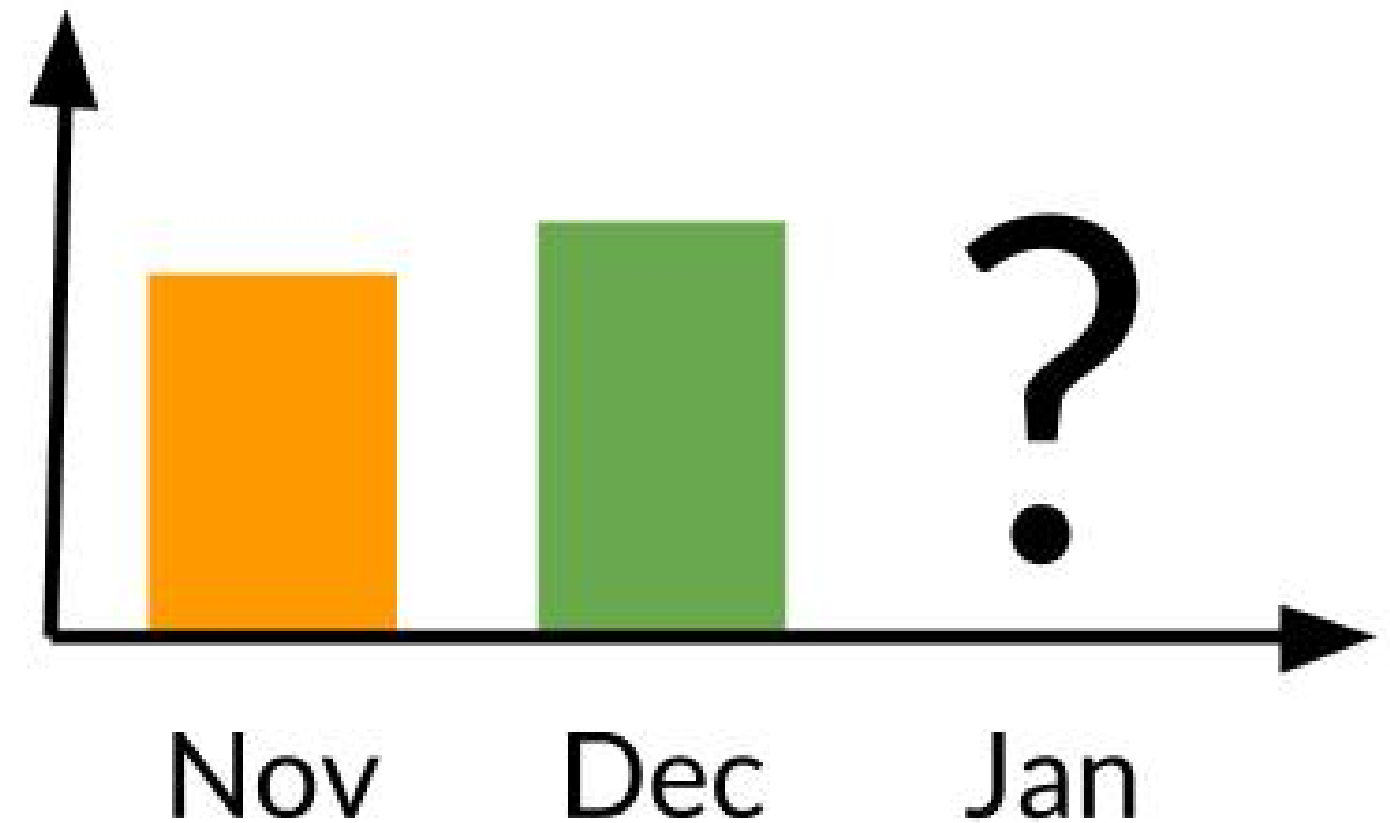
What is an ad hoc request?

- **Not** repeated on a weekly or daily basis
- Can come from many places
 - Product
 - Finance
 - Engineering



Case study: ad campaign performance

The marketing team at your company just wrapped up a new year resolutions ad campaign. Now they want to know how this campaign affected their sales in January.



Marketing asks the Analytics team to provide the total revenue from users who had clicked on one of these ads

Making an ad hoc request

- Be specific
- Include context
- Include a priority level and due date

Handling ad hoc requests

- Ticketing system
 - Ex: Trello, JIRA, Asana
- Require fields
 - Due date
 - Priority

Create a new ticket

Title

Request

We launched a New Year's Resolutions ad campaign in January.

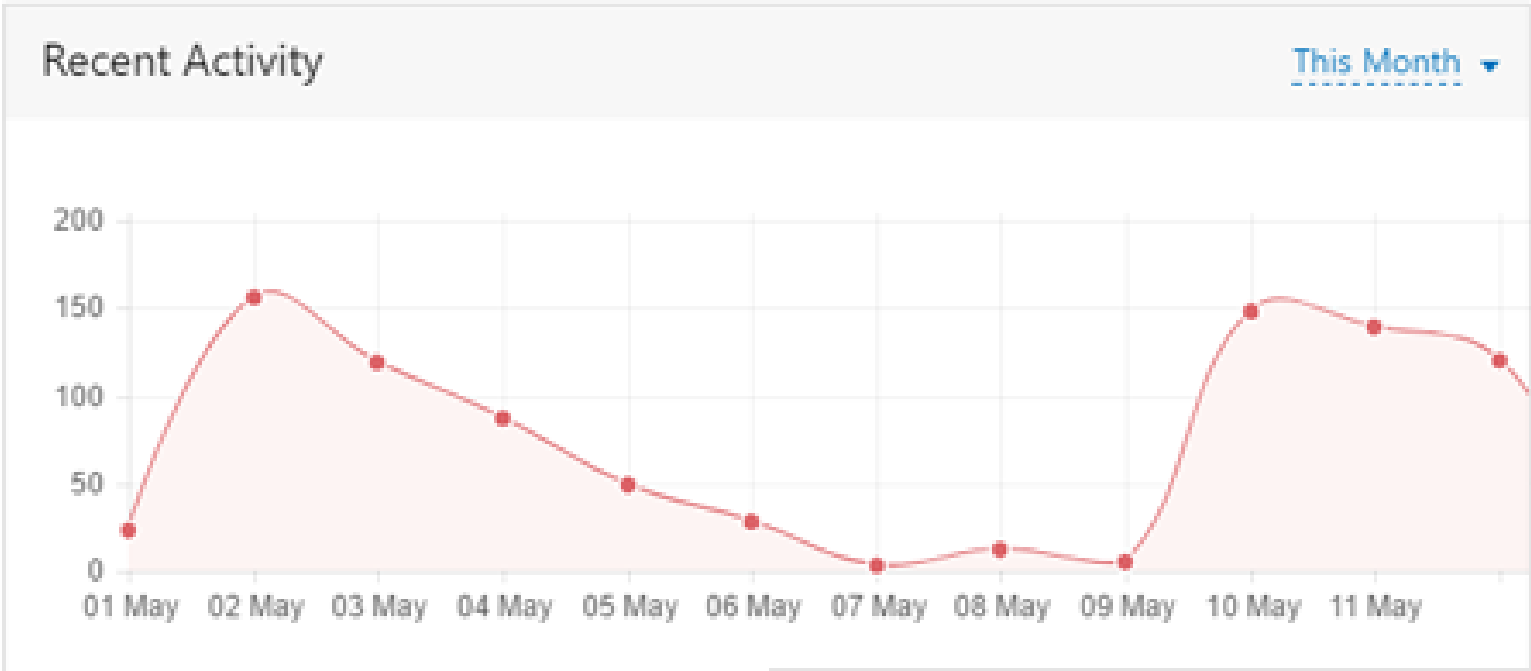
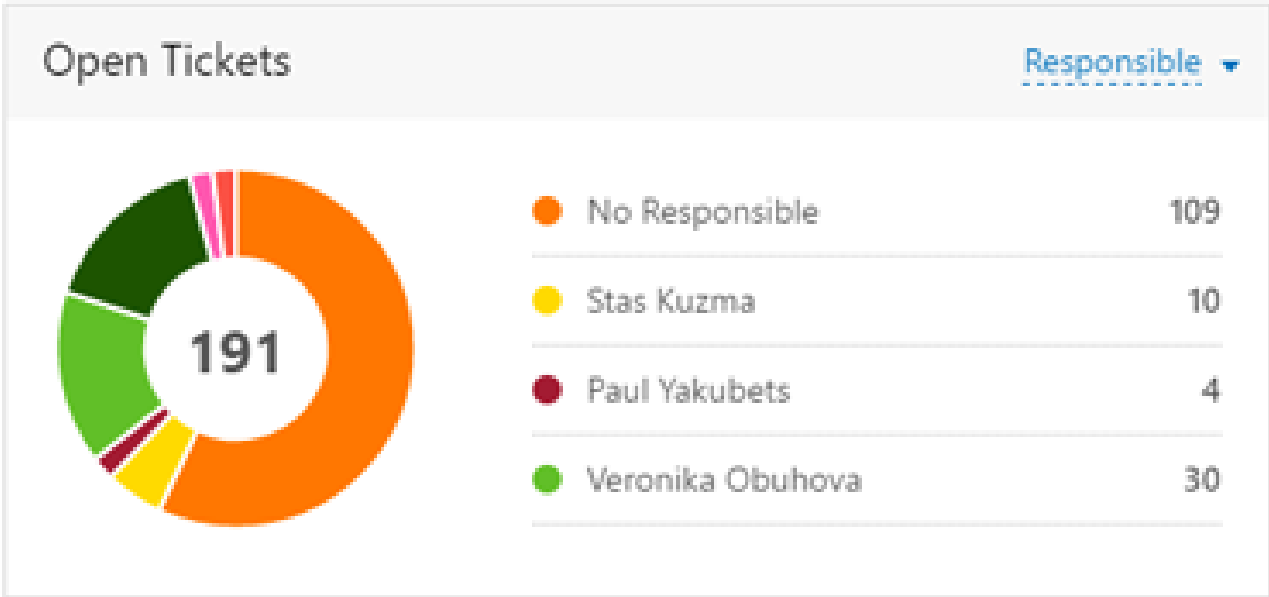
How much revenue was generated by this campaign during that month?

Priority ▼

Due date

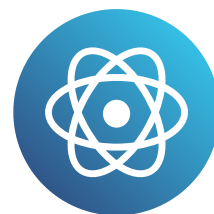
Create

Meta-analysis of Ad Hoc Requests



A/B Testing

DATA SCIENCE FOR BUSINESS LEADERS



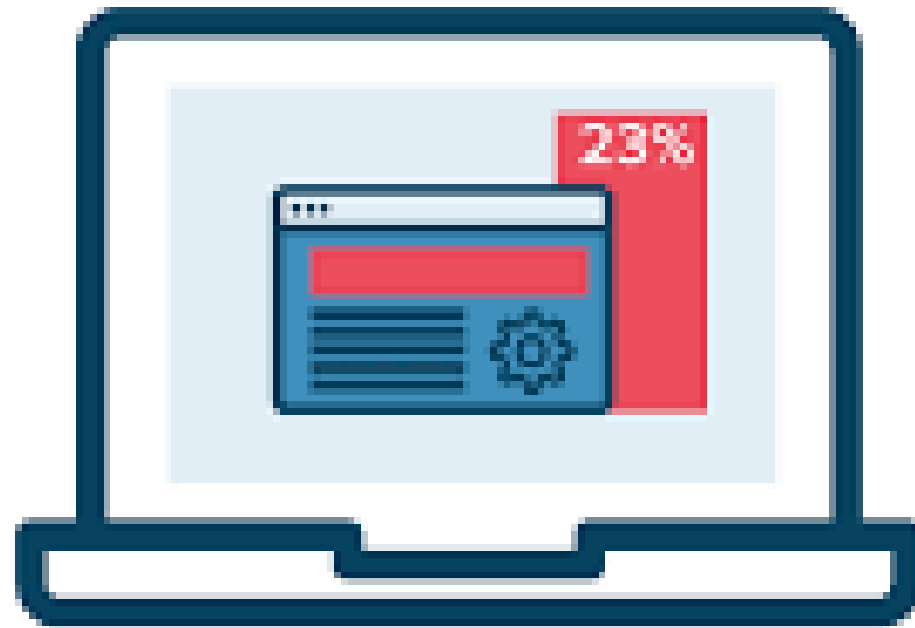
What is A/B Testing?

Powerful
Analytics Tool

It is a type of
experiment for
de-risking
choices between
a few options.

such as, changes
to a website or
wording of email
subjects

A



CONTROL

B



VARIATION

Case study: article headlines

A

Become an expert Data Scientist with this one weird trick!



B

You won't believe these tips for becoming a Data Scientist!

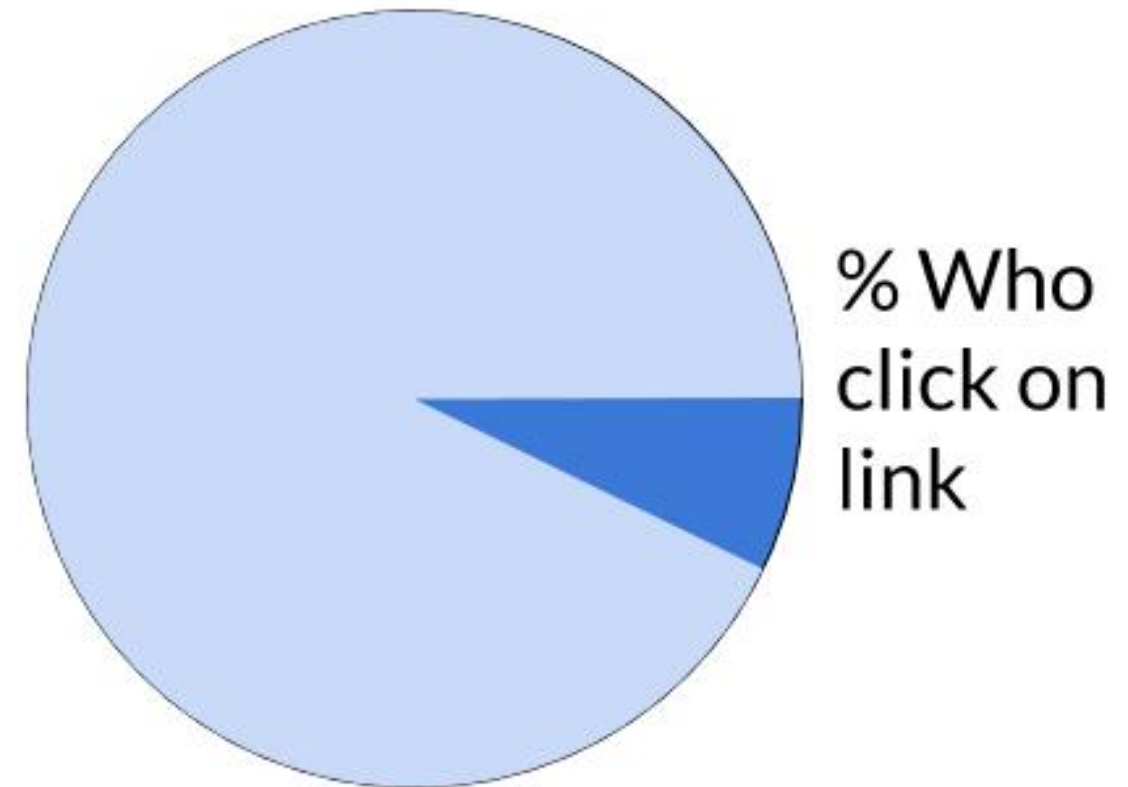


A/B Testing Steps

- Picking a metric to track
- Calculating sample size
- Running the experiment
- Checking for significance

Pick a metric to track

Percentage of people who click on a link with the title of the article

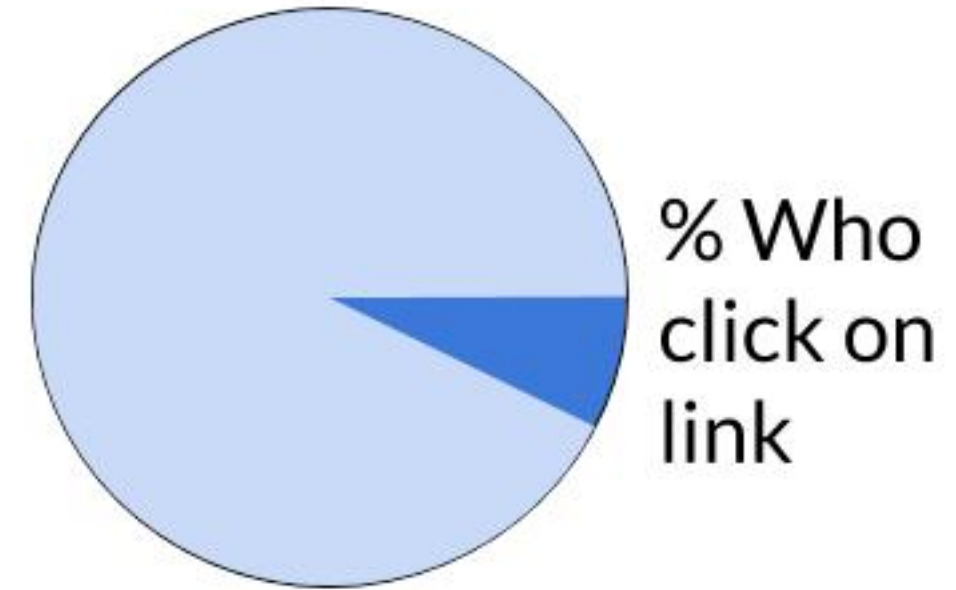
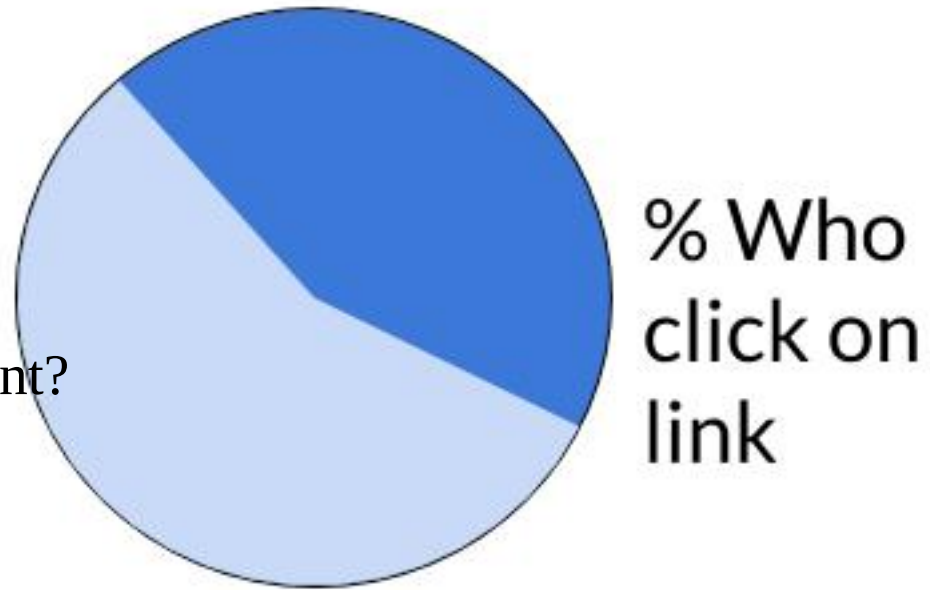


Next Question:

How long to
run an experiment?

Until we reach
a sample size
large enough
that any
difference we
observe is not
due to random chance.

The necessary
sample size
depends on a
baseline metric



In this case,
our baseline
metric is how
often people
generally click
on the link to
our blog.

If it is closer to
50%, we need
a smaller
sample size.

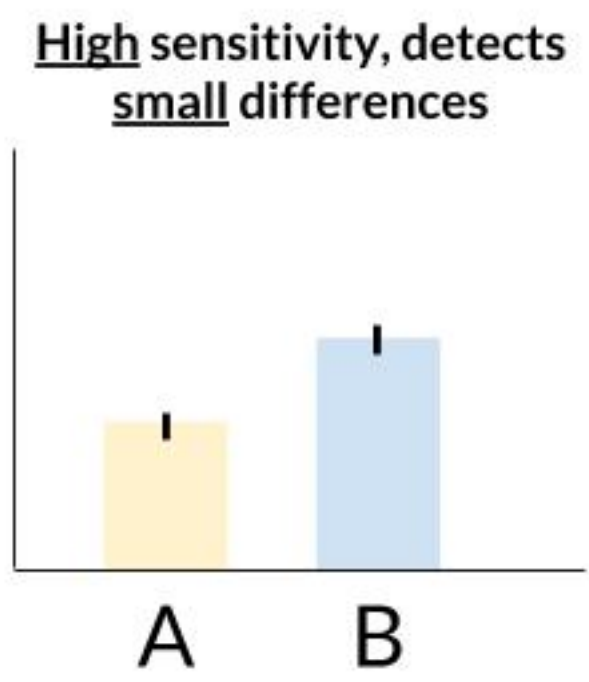
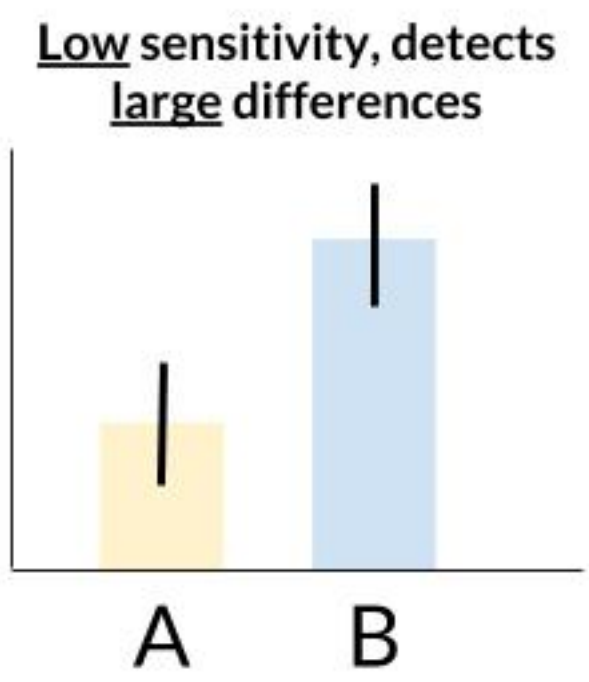
If the rate is
much larger or
much smaller,
we need a
larger sample

The sample size also depends on how sensitive we want our tests to be.

A test sensitivity tells us how small of a change in our metric we are able to detect.

We actually want to optimize the amount of sensitivity that is meaningful for our business problem.

E.g. difference of 0.1 percent in our results

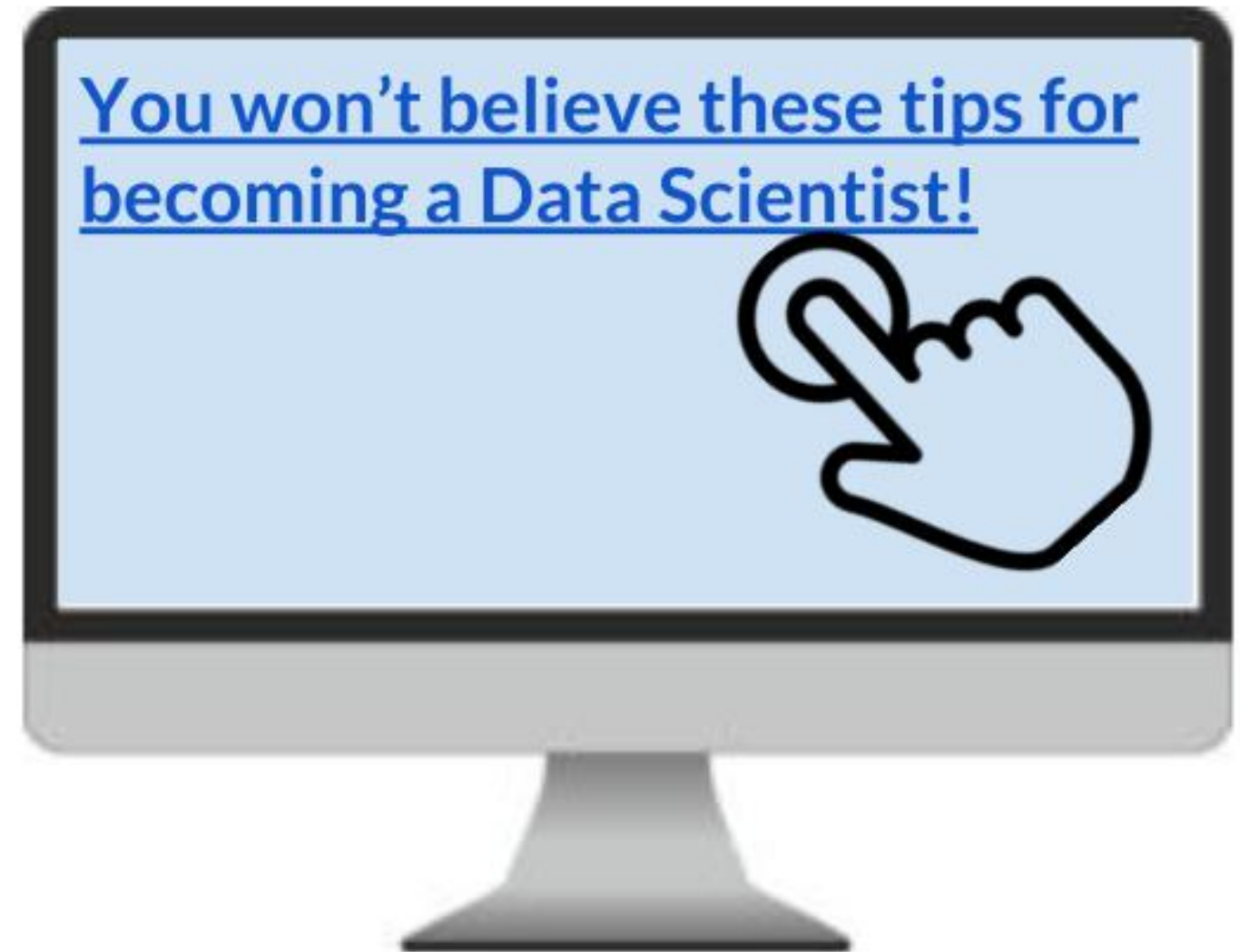


Run your experiment

A



B

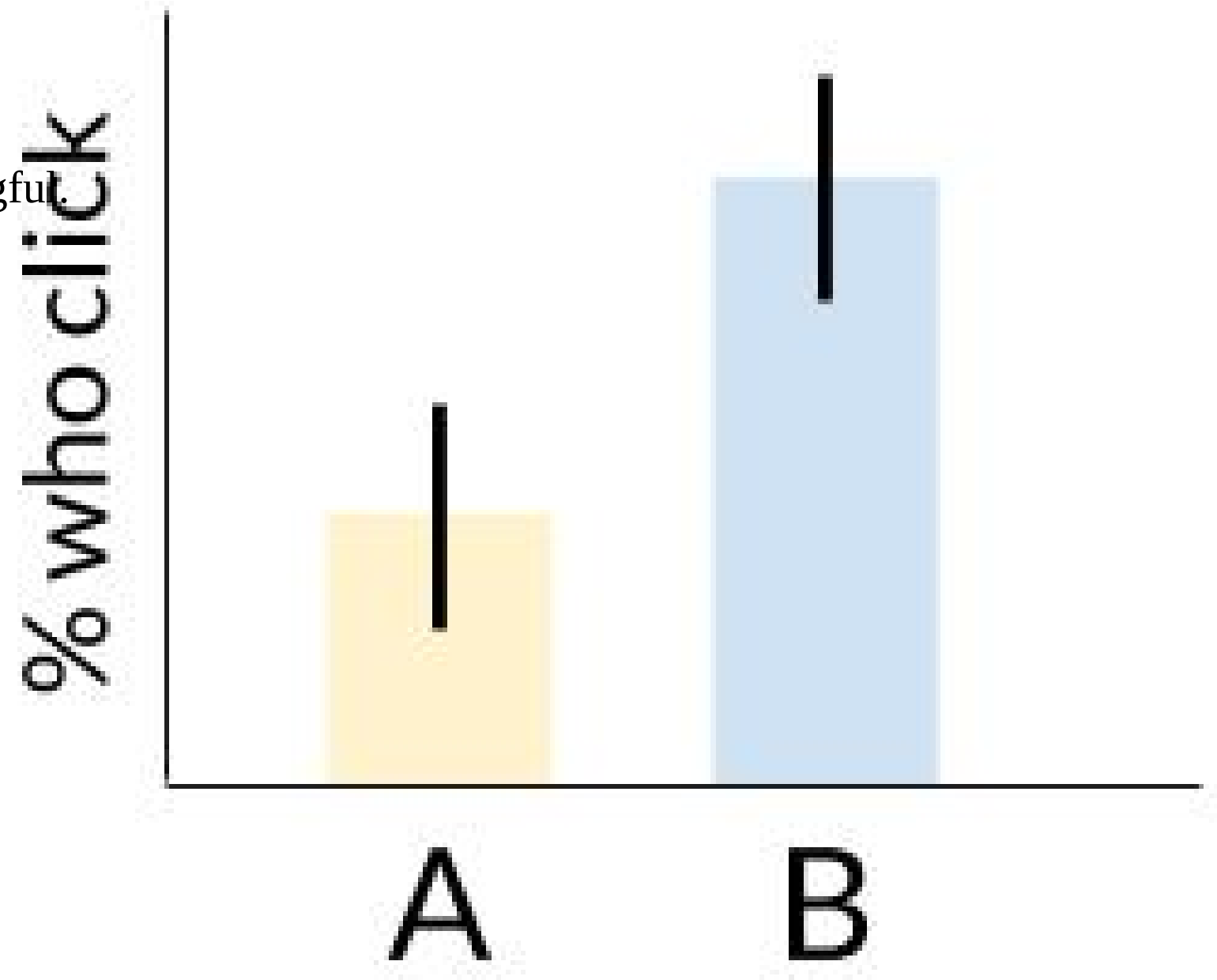


Check for significance

We observe the difference between the 2 titles but how do we know the difference is meaningful.

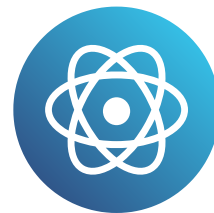
We check by performing a statistical test of significance.

If the results are significant, we can be sure the difference identified can not be due to random chance



Supervised machine learning

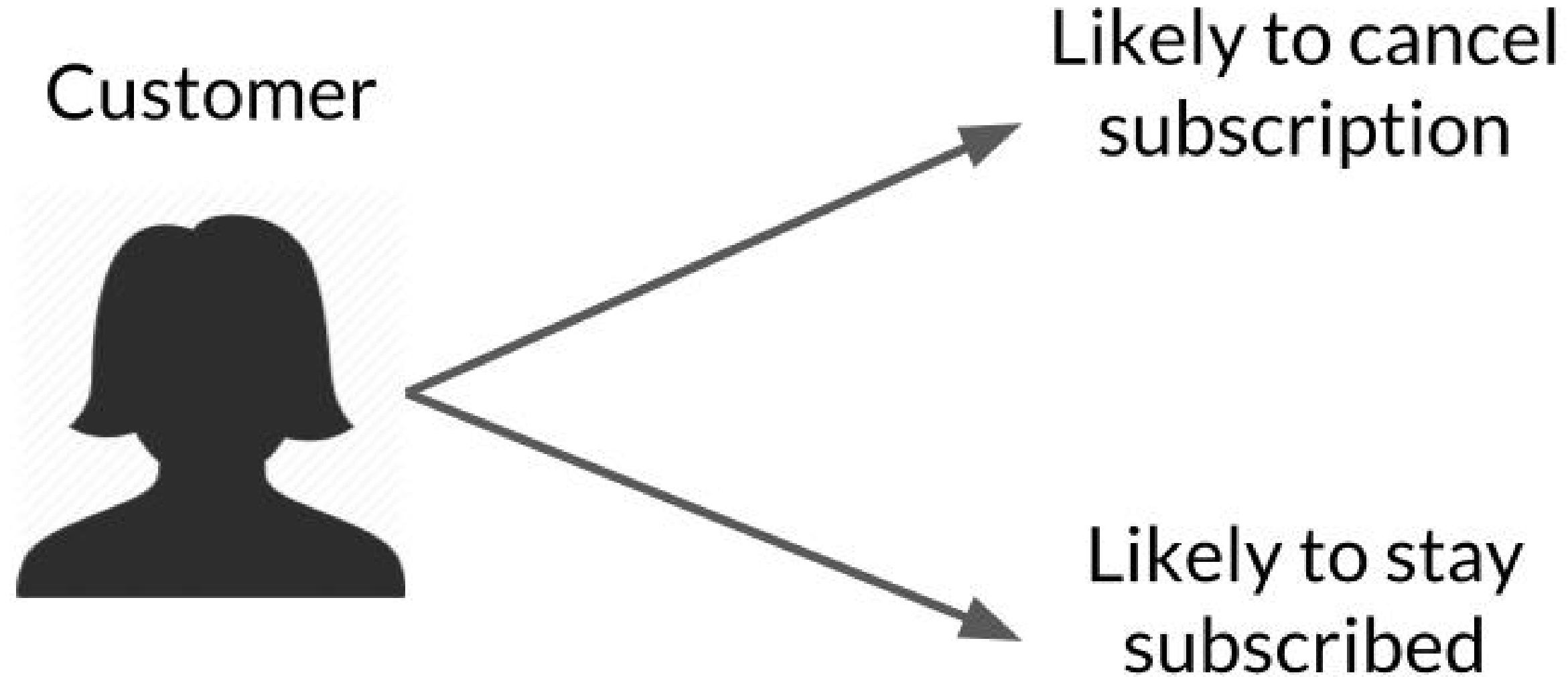
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What is supervised machine learning?

- Machine learning: Predictions from data
- ***Supervised machine learning***: Predictions from data with *labels* and *features*
 - Recommendation systems
 - Email subject optimization
 - Churn prediction

Case study: churn prediction

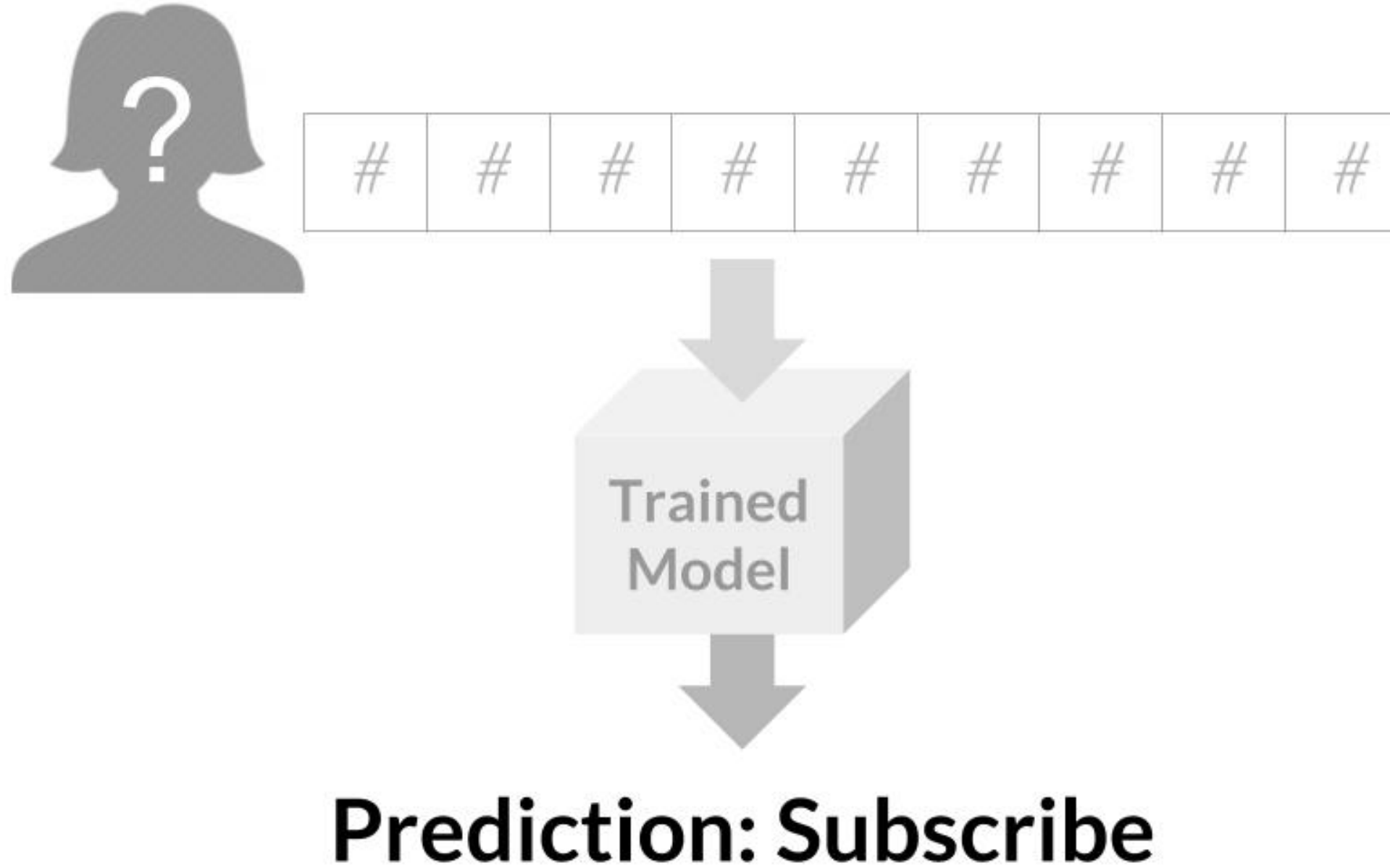


Case study: churn prediction

	Age	Gender	Date of last purchase?	Date of last visit?	Likes cats?	Household \$\$	Location	Number of Kids	Profession	
										churn
										subscribe
										subscribe
										churn
										subscribe
										churn

Features:
Collected customer data

Case study: churn prediction



Model evaluation

Split historical data into training and testing sets

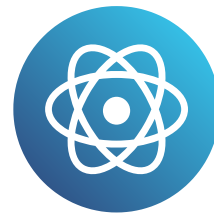


Model evaluation

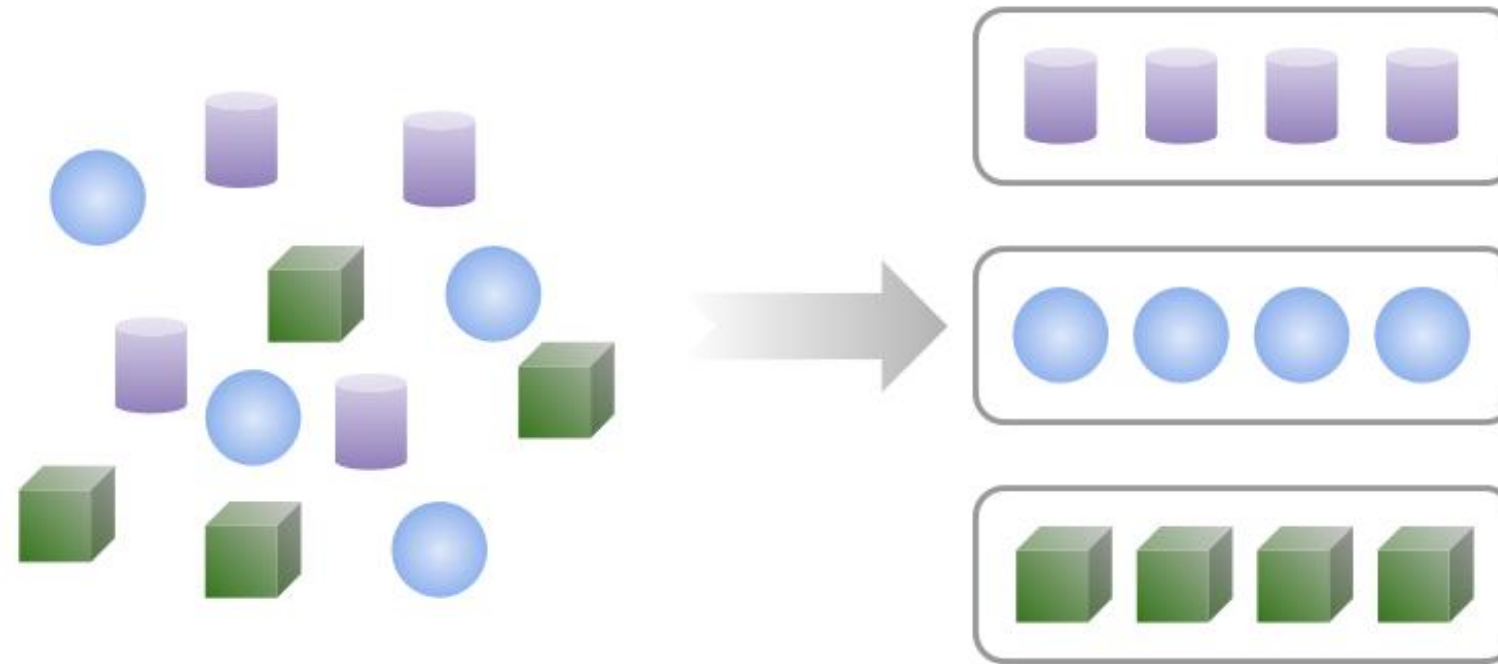
	Prediction	Reality
Churn	0%	3%
Remain	100%	97%

Clustering

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What is clustering?



- Divide data into categories
- Use cases
 - Customer segmentation
 - Image segmentation
 - Anomaly detection

Supervised Machine Learning

	Age	Gender	Date of last purchase?	Date of last visit?	Likes cats?	Household \$\$	Location	Number of Kids	Profession	
										churn
										subscribe
										subscribe
										churn
										subscribe
										churn

Features

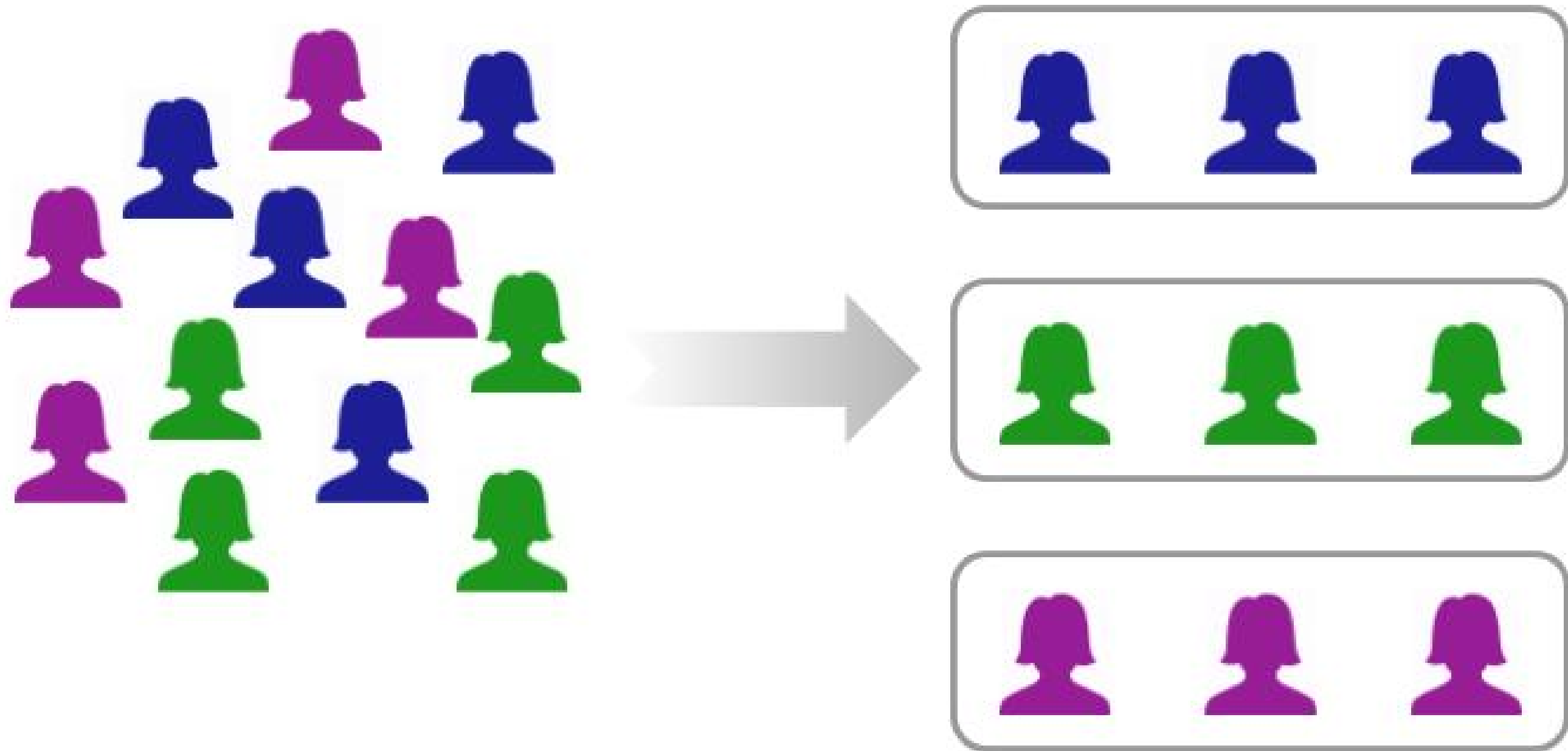
Labels

Unsupervised Machine Learning

	Age	Gender	Date of last purchase?	Date of last visit?	Likes cats?	Household \$\$	Location	Number of Kids	Profession
									
									
									
									
									
									

Features

Case study: customer segmentation



Case study: customer segmentation

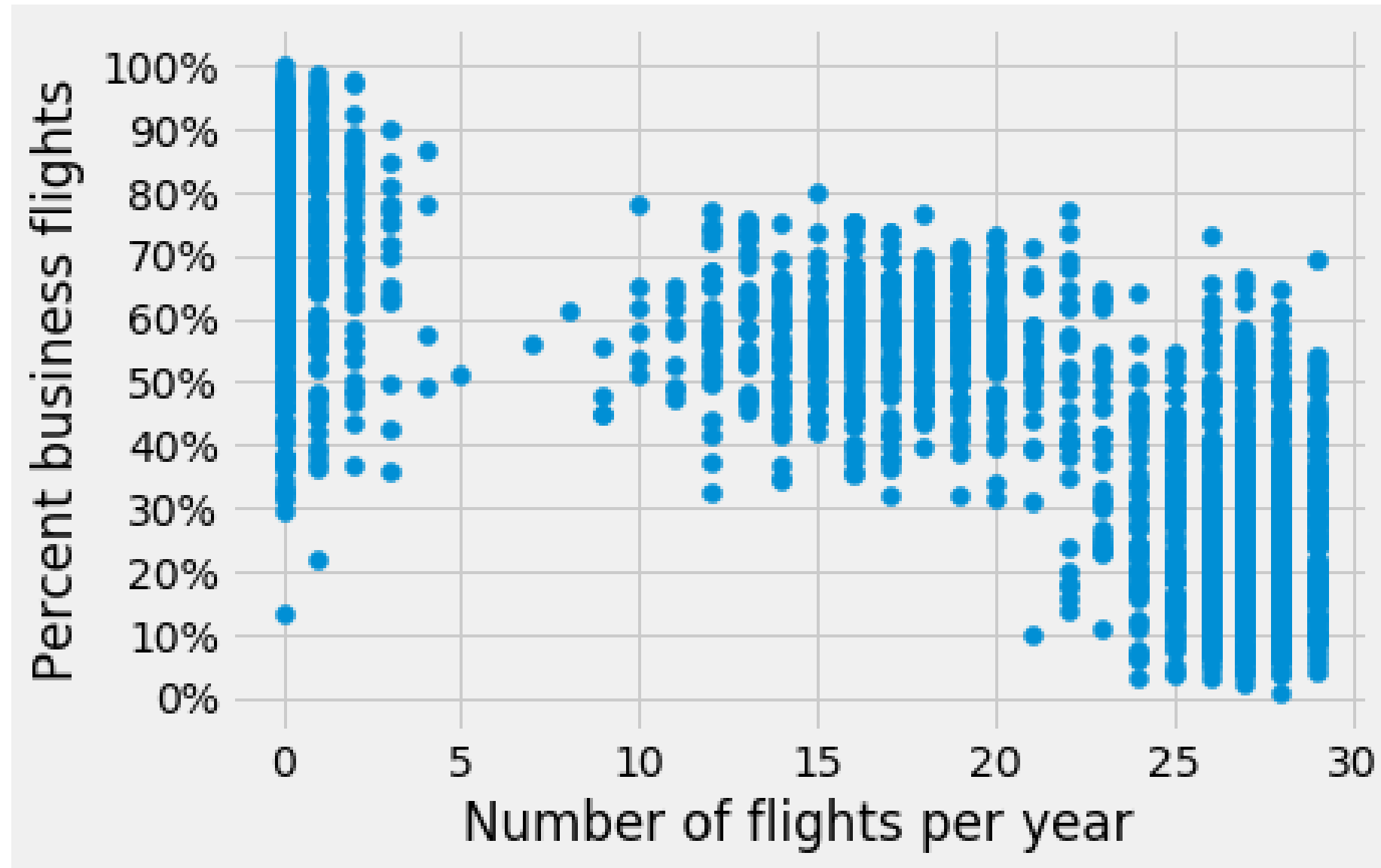
Define features

- Number of flights in the past year
- Percent international
- Advanced planning
- Percent business class

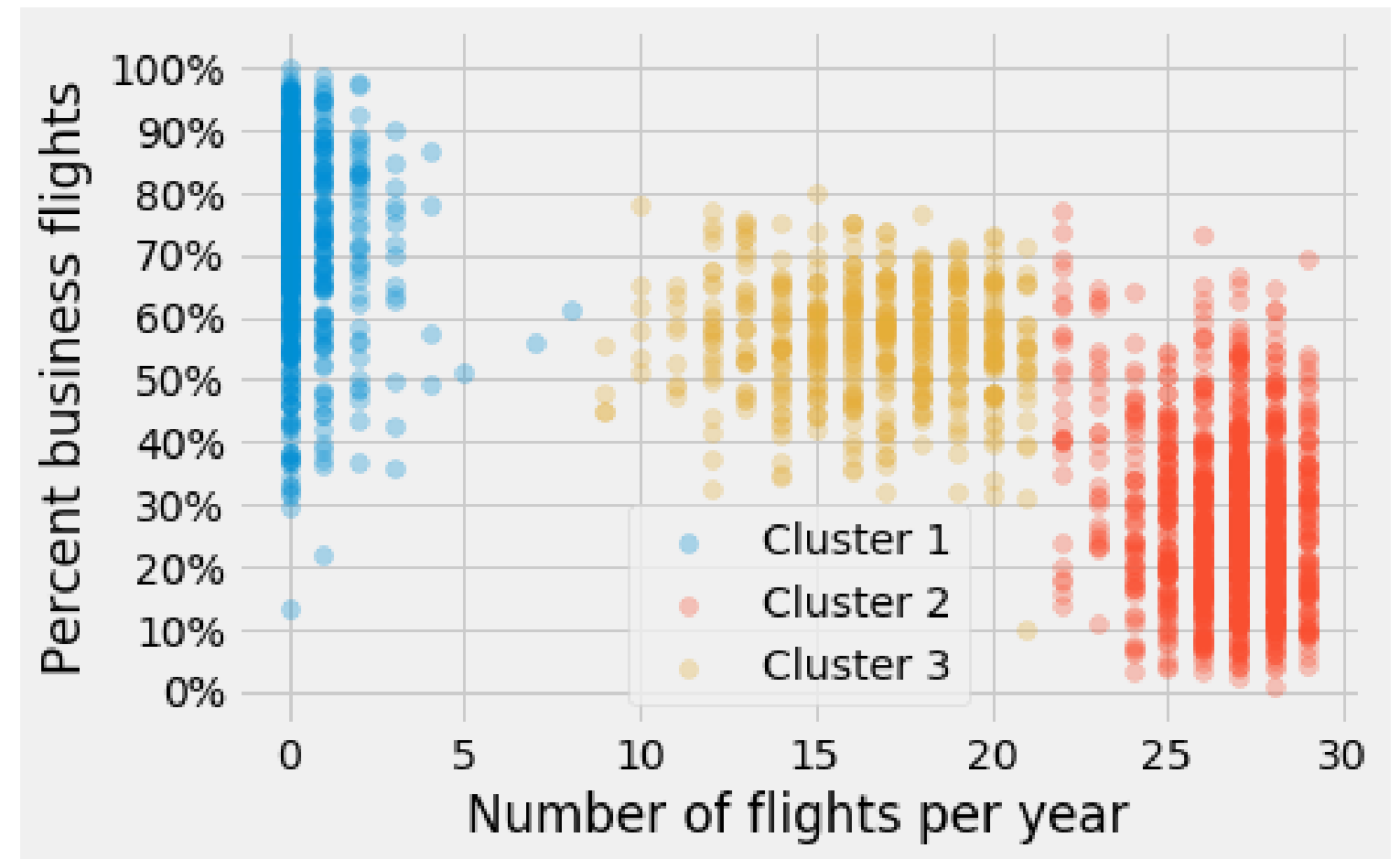
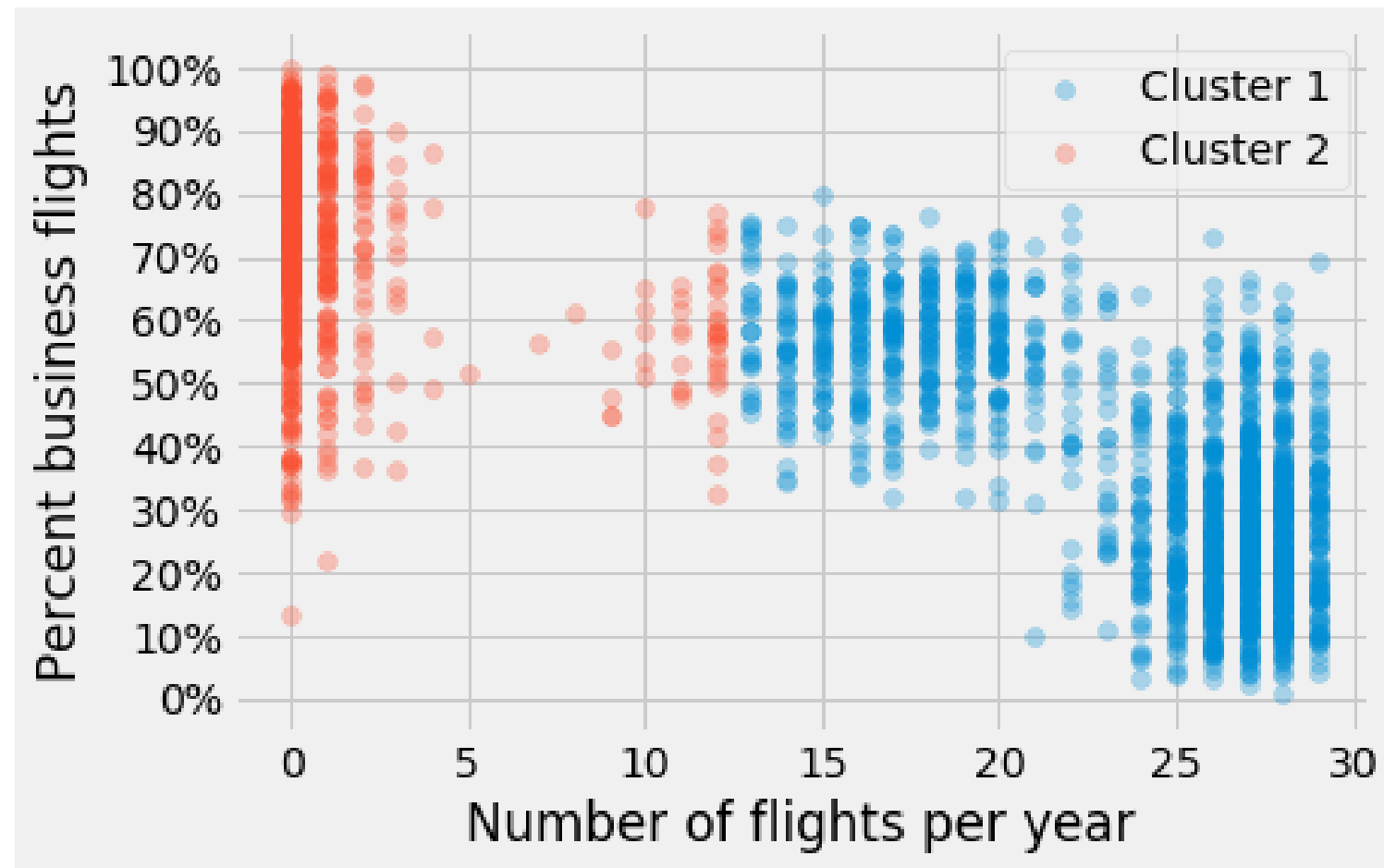


Case study: customer segmentation

- Define number of clusters

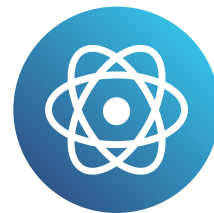


Case study: customer segmentation

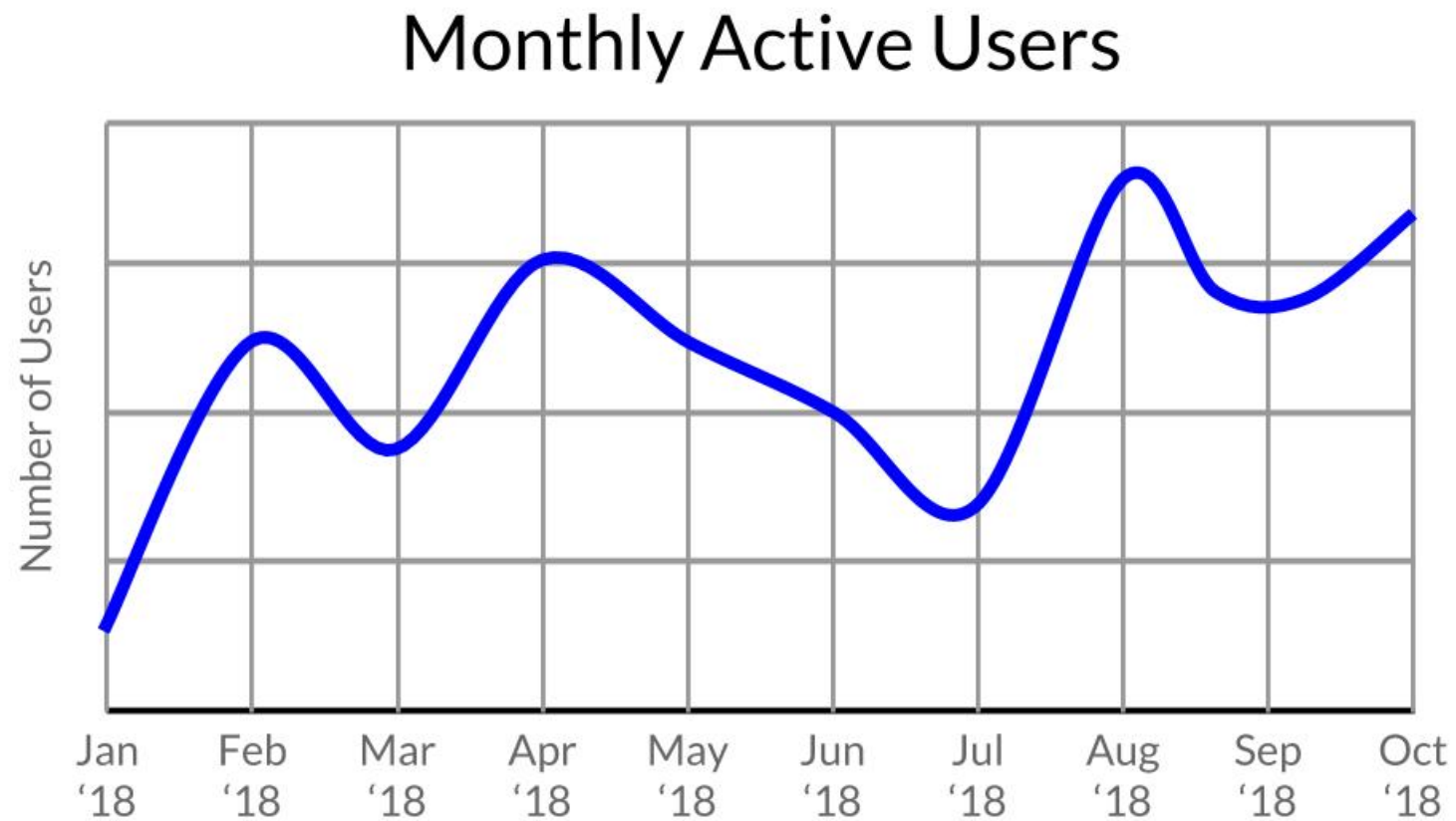


Special topics in Machine Learning

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Time series forecasting



- Time is a feature
- Accounts for weekly, monthly, or yearly trends

Seasonality

- **Weekly:** Lower television viewership on Fridays
- **Monthly:** Higher spending at end of pay periods
- **Yearly:** Less ice cream in the winter



Natural Language Processing

- Dataset is text
 - Customer reviews
 - Tweets
 - Medical records
 - Email subjects
- Possible uses
 - Classifying sentiment
 - Clustering medical records



Word counts

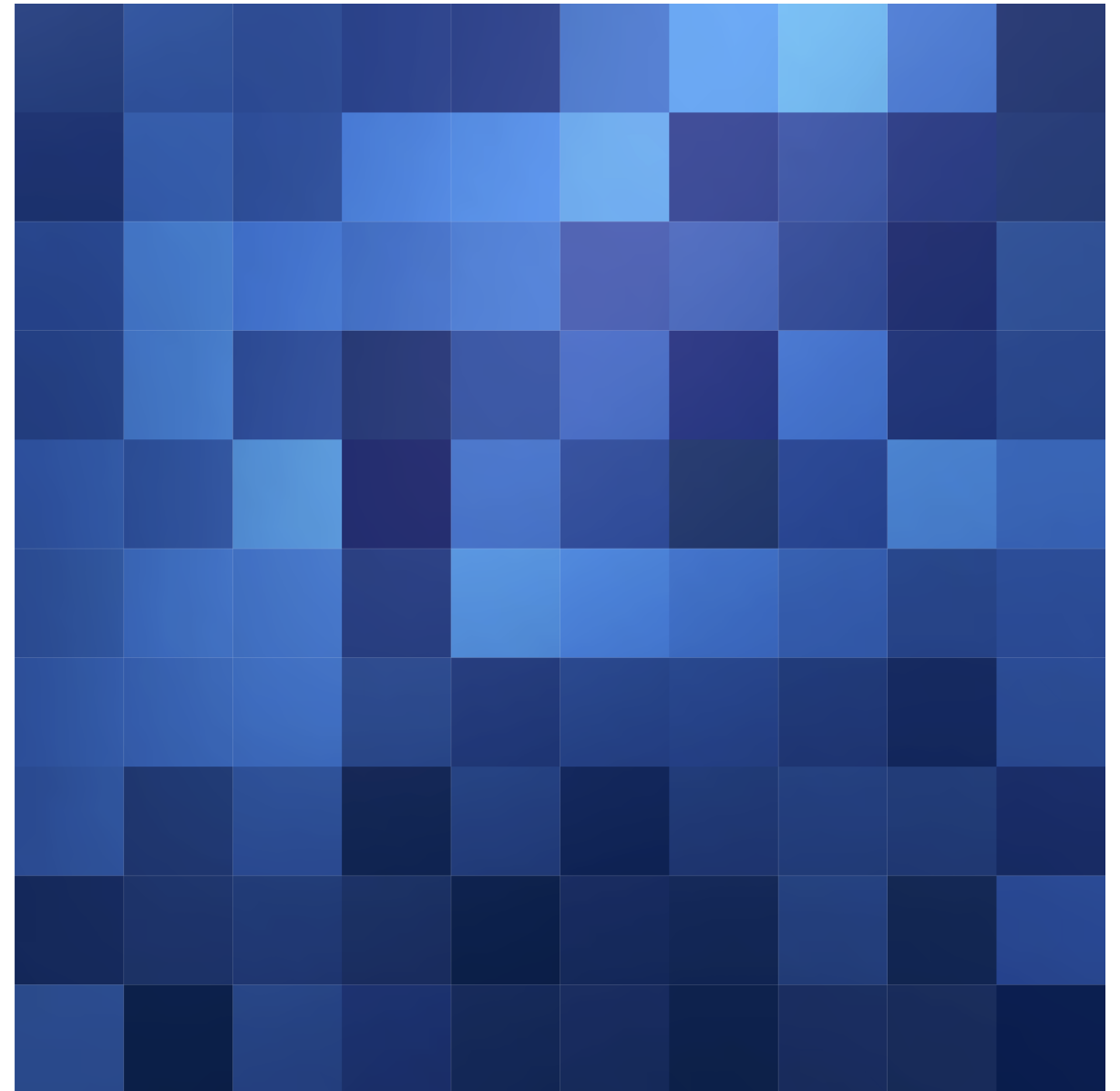
Sentence	Texans	Giants	football	great
The Texans are a great football team.	1	0	1	1
The Giants are a great football team.	0	1	1	1

Problems with word counts: negation

Sentence	Texans	Giants	football	great	not
The Giants are a great football team.	0	1	1	1	0
The Giants are not a great football team.	0	1	1	1	1

Word counts and synonyms

- Word counts don't help us consider synonyms
- Example: "blue"
 - "sky-blue"
 - "aqua"
 - "cerulean"
- Want to group as a single feature



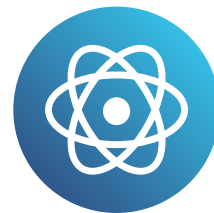
Word embeddings

- Create features that group similar words
- Features have a mathematical meaning:

`king - man + woman = queen`

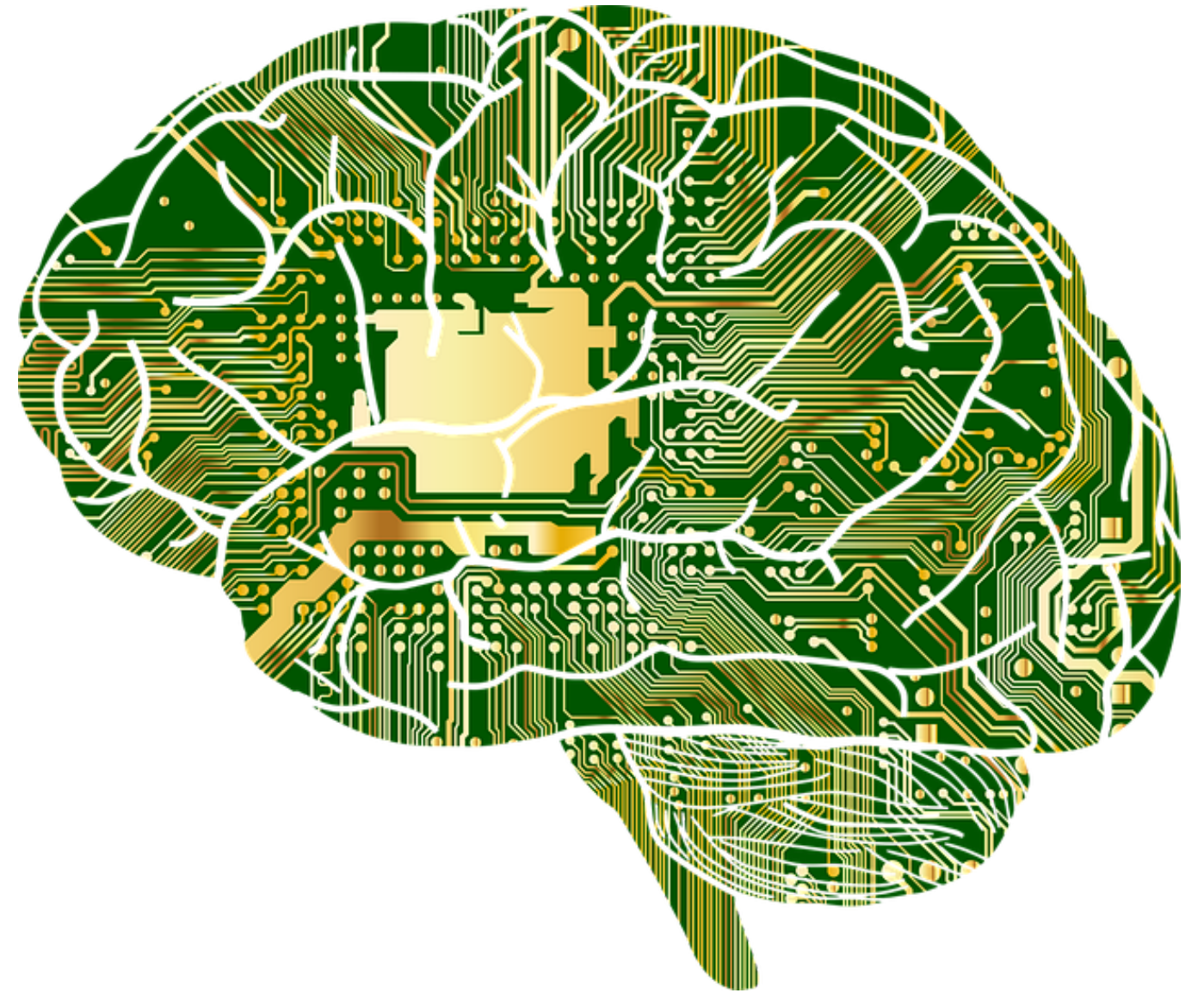
Deep Learning and Explainable AI

DATA SCIENCE FOR BUSINESS LEADERS



What is Deep Learning?

- AKA: Neural Networks or Neural Nets
- Special area of Machine Learning
- Requires more data
- Best when inputs that are images or text



Explainable AI

Deep Learning	Explainable AI
Highly accurate predictions	Understandable by humans
Better for "What?"	Better for "Why?"

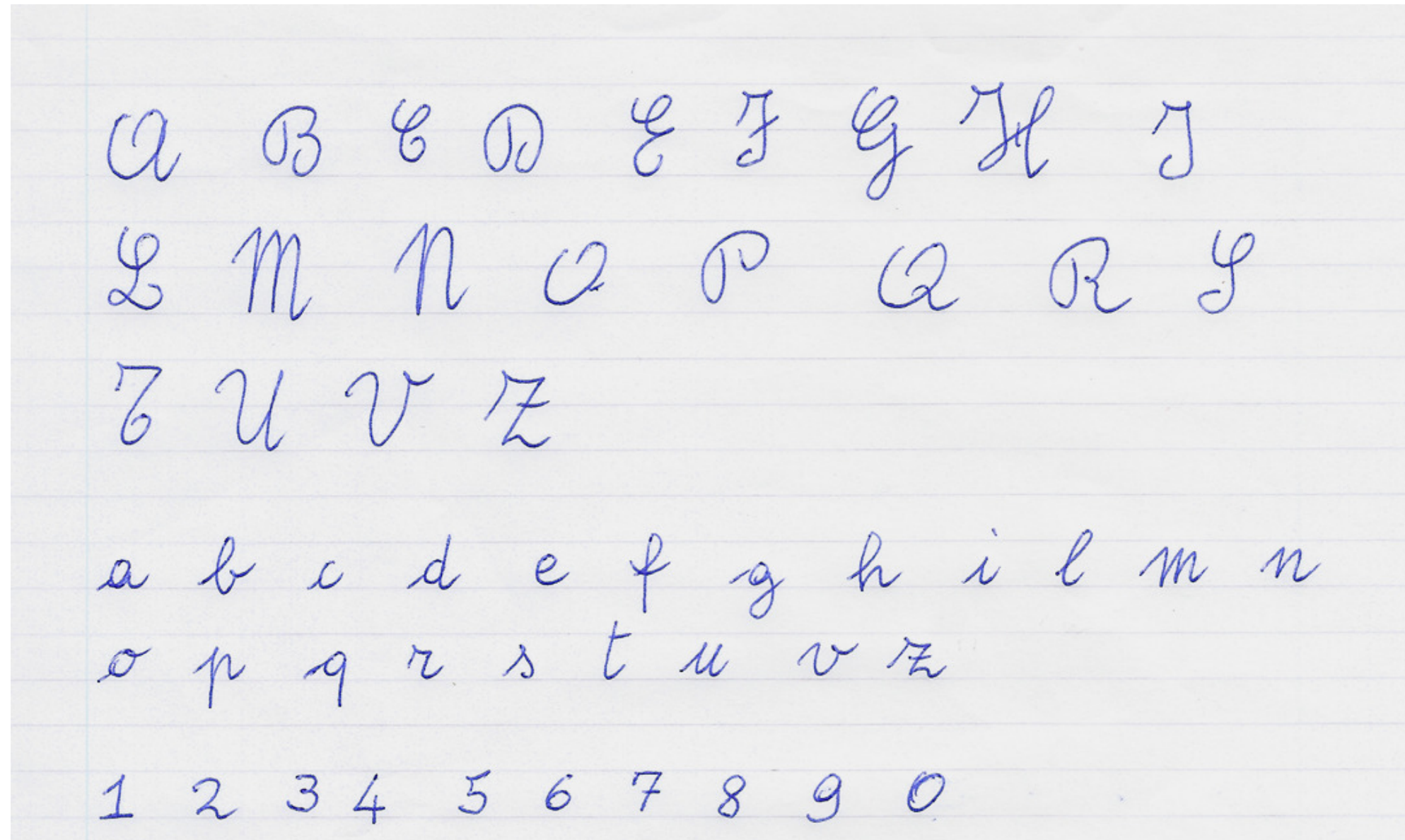
Case Study: Explainable AI

1. **Prediction:** What a customer is likely to do
2. **Explanation:** Why a customer is likely to do it



Case Study: Inexplicable AI

Prediction only: Which letter is this likely to be?



When to use Deep Learning

1. Is the training data complex?
2. Do we have a very large amount of training data?
3. Does the model need to be predictive or explanatory?

Programming & Python

“In the next few decades, you will have to know a programming language in order to communicate with people,” Broadway said

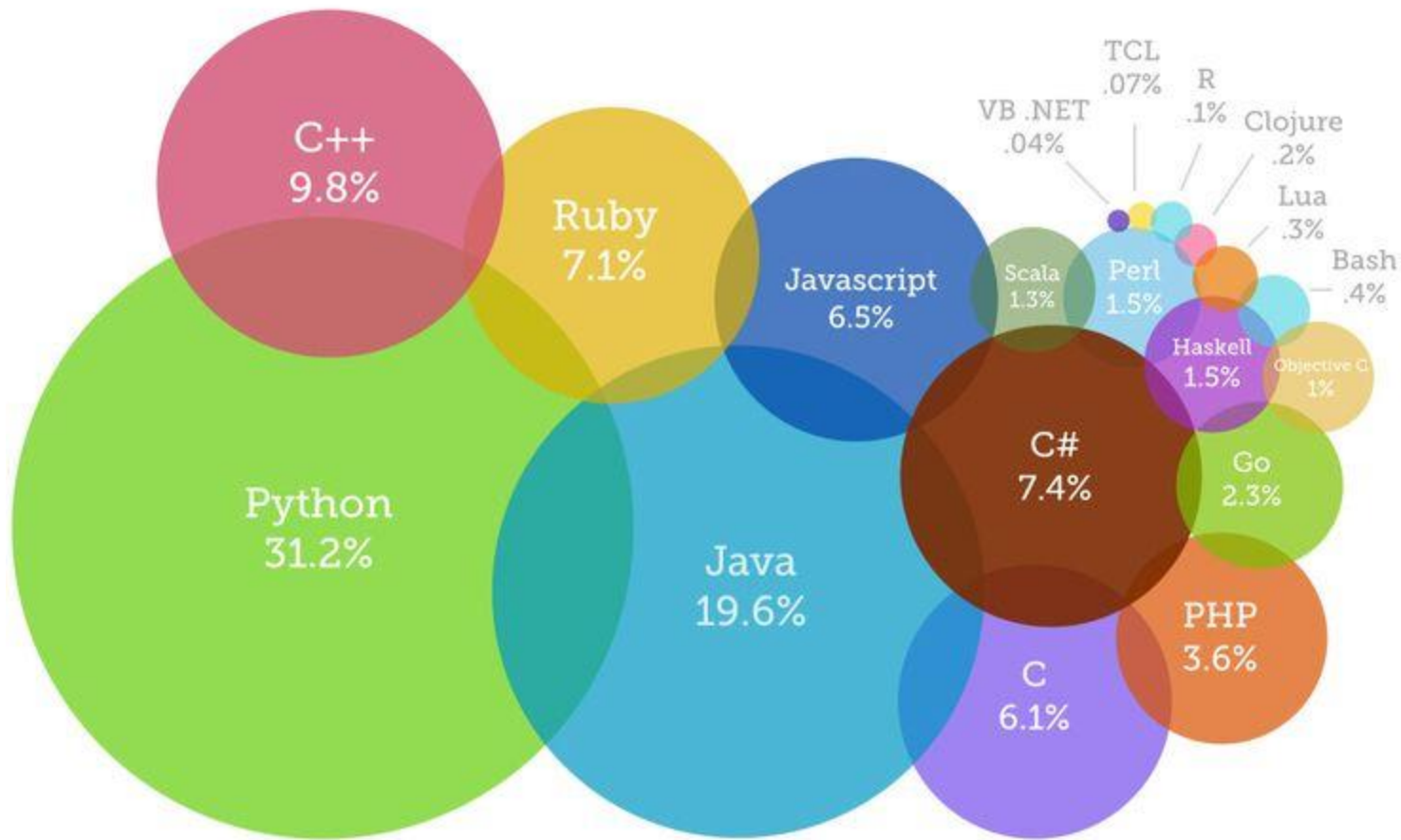
Python

- Employers' desire for expertise in programming and data analysis
- Programming spans a variety of functions and industries: finance, operations, marketing, healthcare, and so on.
- Even if the MBAs won't be working with the programs themselves, they will probably be managing a team who does work with them.
- "Employers across industries have access to an ever-growing amount of data in this highly digitized and quantified environment we all live in," said Emily Anderson, Director of the Career Management Center.
- "This means the more exposure, education, and analytical skill development a current student can gain while they are in the program, the more valuable they will be to their new potential employers."

Python

- While many students start off working in data interfaces that provide some level of organization and interpretation, learning how to extract what information they need from raw data is another critical skill.
- We're going to need to not have an interface if we really want to get to analysis and data that makes sense for us. We're going to have to create the queries (ourselves)
- This training also equips students to think critically about data that's presented to them by others, a key skill for anyone working in the business world. "There's so many ways to manipulate data in how you show it to people."

Programming Languages



Python

Organizations using Python :

- 1) Google(Components of Google spider and Search Engine)
- 2) Yahoo(Maps)
- 3) YouTube
- 4) Mozilla
- 5) Dropbox
- 6) Microsoft
- 7) Cisco
- 8) Spotify
- 9) Quora

Python Language Introduction

- Python is a widely used general-purpose, high level programming language. It was initially designed by Guido van Rossum in 1991 and developed by Python Software Foundation. It was mainly developed for emphasis on code readability, and its syntax allows programmers to express concepts in fewer lines of code.
- Python is a programming language that lets you work quickly and integrate systems more efficiently.
- There are two major Python versions- **Python 2 and Python 3**. Both are quite different.

Beginning with Python programming

1. Install Python

Visit *python.org* and click on Community Edition to Download Python setup and run to install on recommended settings

2. Finding an Interpreter:

Before we start Python programming, we need to have an interpreter to interpret and run our programs. There are certain online interpreters like <https://ide.geeksforgeeks.org/>, <http://ideone.com/> or <http://codepad.org/> that can be used to start Python without installing an interpreter.

- **Windows:** There are many interpreters available freely to run Python scripts like IDLE (Integrated Development Environment) which is installed when you install the python software from <http://python.org/> or IntelliJ IDEA or PyCharm
- **Linux:** For Linux, Python comes bundled with the linux.

Reason for increasing popularity

- Emphasis on code readability, shorter codes, ease of writing
- Programmers can express logical concepts in fewer lines of code in comparison to languages such as C++ or Java.
- Python supports multiple programming paradigms, like object-oriented, imperative and functional programming or procedural.
- There exists inbuilt functions for almost all of the frequently used concepts.
- Philosophy is “**Simplicity is the best**”.
- Libraries are there to ease your work
- A very strong community of learners and experts is available on different platform to support your work.

Language Features

- **Interpreted**

- There are no separate compilation and execution steps like C and C++.
- Directly *run* the program from the source code.
- Internally, Python converts the source code into an intermediate form called bytecodes which is then translated into native language of specific computer to run it.
- No need to worry about linking and loading with libraries, etc as it is done automatically

- **Platform Independent**

- Python programs can be developed and executed on multiple operating system platforms.
- Python can be used on Linux, Windows, Macintosh, Solaris and many more.

- **Free and Open Source; Redistributable**

- **High-level Language**

- In Python, no need to take care about low-level details such as managing the memory used by the program.

Language Features

- **Simple**

- Closer to English language; Easy to Learn
- More emphasis on the solution to the problem rather than the syntax

- **Embeddable**

- Python can be used within C/C++ program to give scripting capabilities for the program's users.

- **Robust:**

- Exceptional handling features
- Memory management techniques in built

- **Rich Library Support**

- The Python Standard Library is vary vast.
- Known as the “**batteries included**” philosophy of Python ;It can help do various things involving regular expressions, documentation generation, unit testing, threading, databases, web browsers, CGI, email, XML, HTML, WAV files, cryptography, GUI and many more.
- Besides the standard library, there are various other high-quality libraries such as the Python Imaging Library which is an amazingly simple image manipulation library.

Pros & Cons

Pros:

- Ease of use
- Multi-paradigm Approach

Cons:

- Slow speed of execution compared to C,C++
- Absence from mobile computing and browsers
- For the C,C++ programmers switching to python can be irritating as the language requires proper indentation of code. Certain variable names commonly used like sum are functions in python. So C, C++ programmers have to look out for these.

Industrial Importance

- Most of the companies are now looking for candidates who know about Python Programming. Those having the knowledge of python may have more chances of impressing the interviewing panel. So I would suggest that beginners should start learning python and excel in it.

Applications of Python

Web Development

Python can be used to make web-applications at a rapid rate. Why is that? It is because of the frameworks Python uses to create these applications. There is *common-backend logic* that goes into making these frameworks and a number of libraries that can help integrate protocols such as HTTPS, FTP, SSL etc. and even help in the processing of HTML, E-Mail and so much more.

Some of the most well-known frameworks are Django and Flask. Why use a framework? The ***security, scalability, convenience*** that they provide is unparalleled compared to starting the development of a website from scratch.

Web Scraping Applications

Python can be used to pull a large amount of data from websites which can then be helpful in various real-world processes such as price comparison, job listings, research and development and much more. Python has a library called BeautifulSoup which can be used to pull such data and be used accordingly

Applications of Python

Data Analysis in Data Science

Data is money if you know how to extract relevant information which can help you take calculated risks and increase profits. You study the data you have, perform operations and extract the information required. Libraries such as Pandas, NumPy help you in extracting information.

You can even visualize the data libraries such as Matplotlib, Seaborn, which are helpful in plotting graphs and much more. This is what Python offers you to become a Data Scientist

Machine Learning

Applications include Face recognition, voice recognition and recommendation systems. A few other uses may include Personal assistants like Siri and Alexa, Refined Travel Estimates on Google Maps, Product Recommendation Systems, Analysis of customer satisfaction, Research on what factors will impact household pricing and Predict stocks. Machine Learning and Artificial Intelligence are the talks of the town as they yield the most promising careers for the future. The programming language that mostly everyone chooses? It's Python. Why? Support for these domains with the **libraries** that exist already such as Pandas, Scikit-Learn, NumPy and so many more. Learn the algorithm, use the library and you have your solution to the problem.

Applications of Python

Scripting

Designing small programs to automate repetitive tasks e.g. identifying or labeling certain emails, getting data from websites and compiling them with visualizations.

Game Development

Python is also used in the development of interactive games. PyGames is a library that can be used. Games such as Civilization-IV, Disney's Toontown Online, Vega Strike etc. have been built using Python.

Desktop Applications

Python can be used to program desktop applications. It provides the Tkinter library that can be used to develop user interfaces. There are some other useful toolkits such as the wxWidgets, Kivy, PYQT that can be used to create applications on several platforms.

You can start out with creating simple applications such as Calculators, To-Do apps and go ahead and create much more complicated applications.

A few Resources to move further

- Cognitiveclass.ai

An IBM initiative for Data Science

- Edx

Online courses from various institutes around the world. A few are completely free, others provide knowledge for free and charge for a certificate